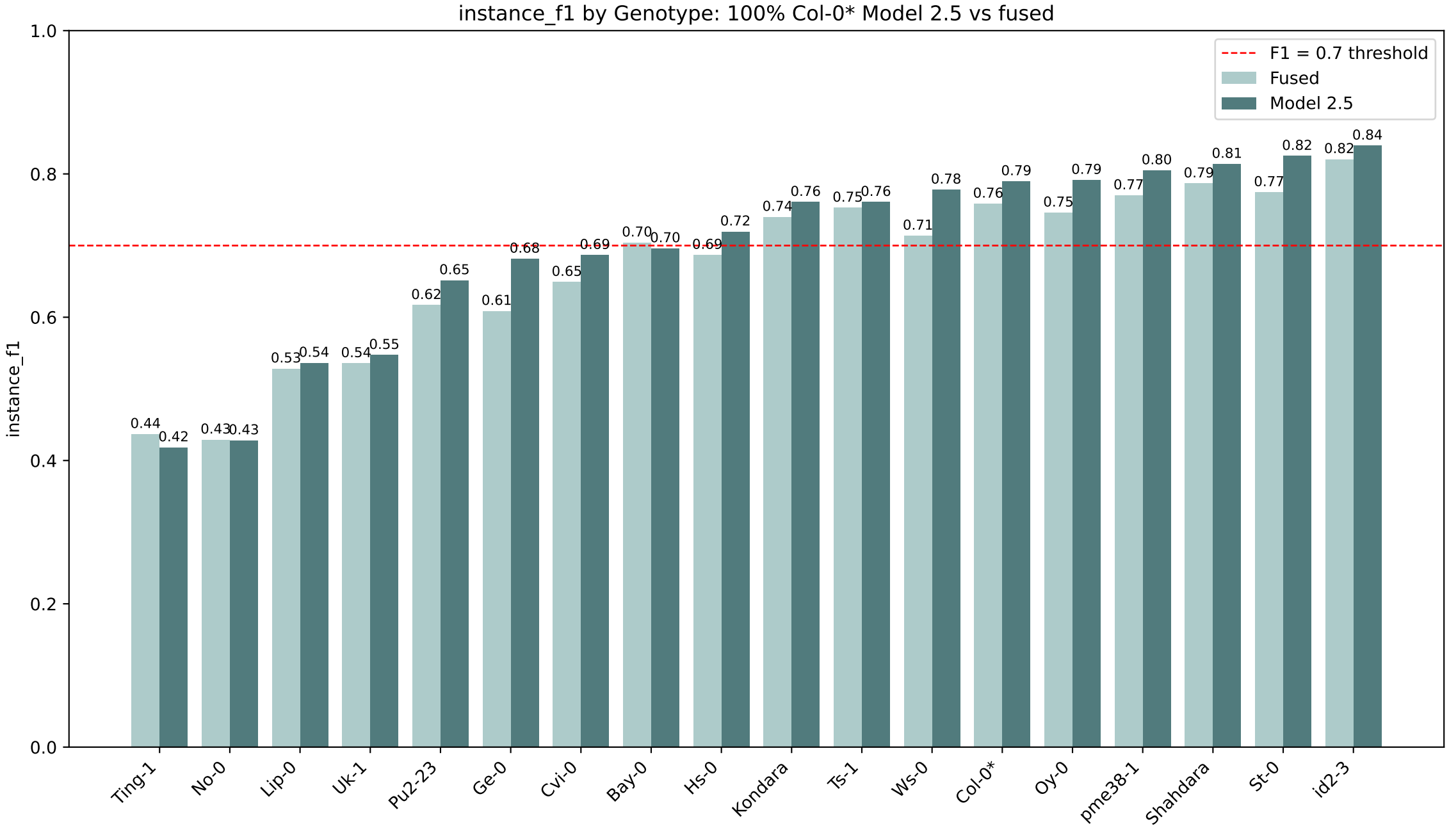
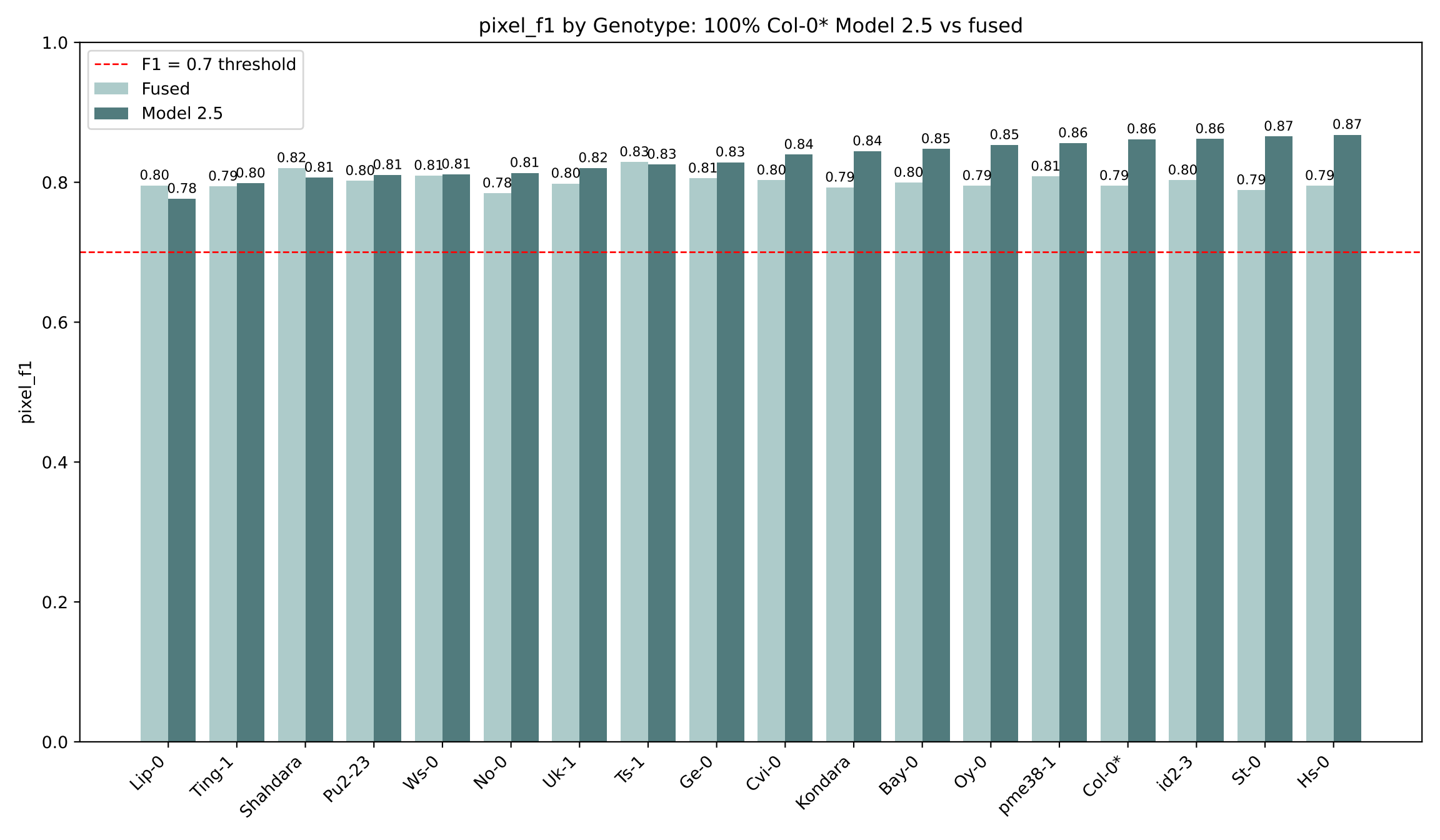
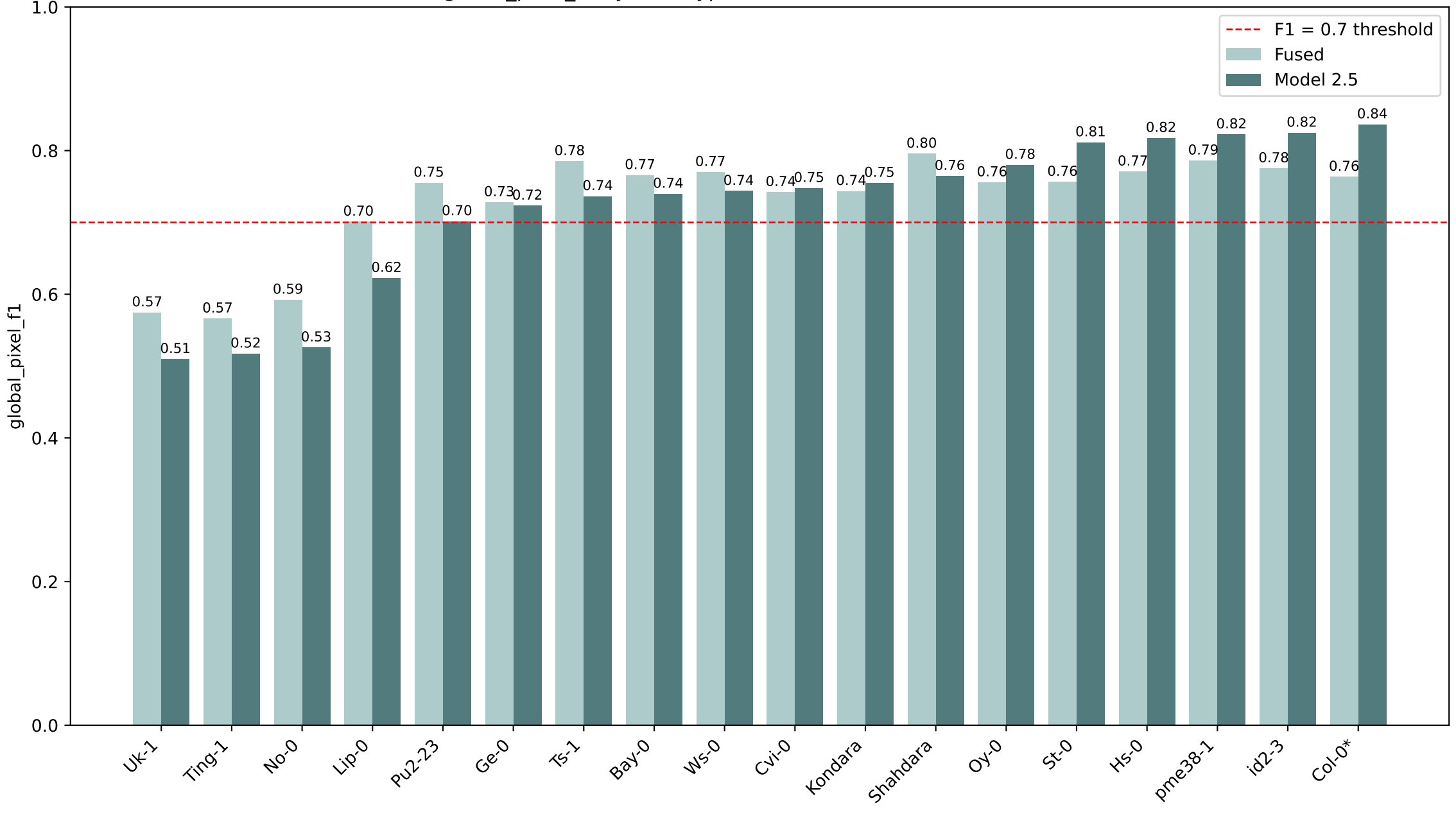
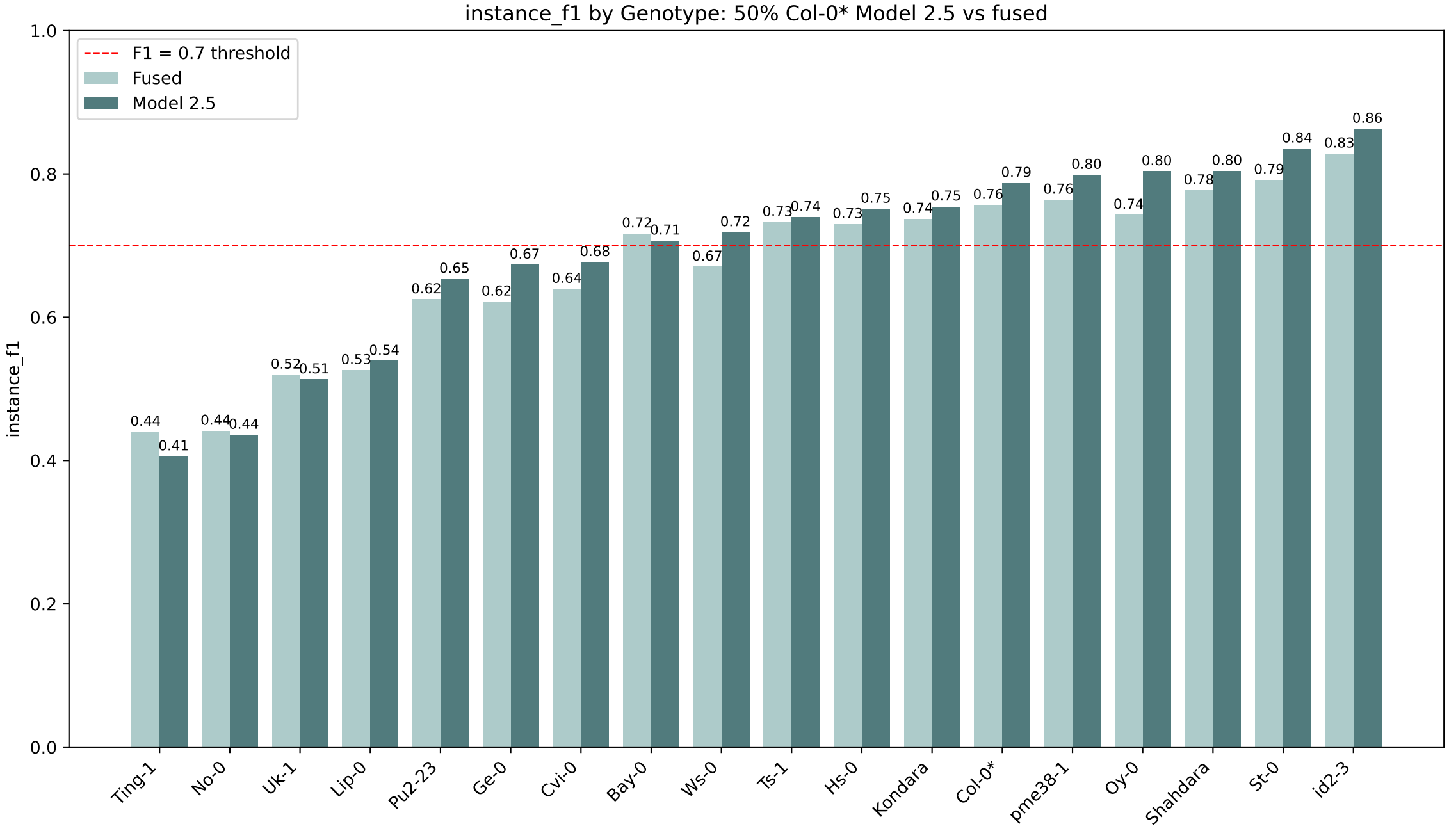
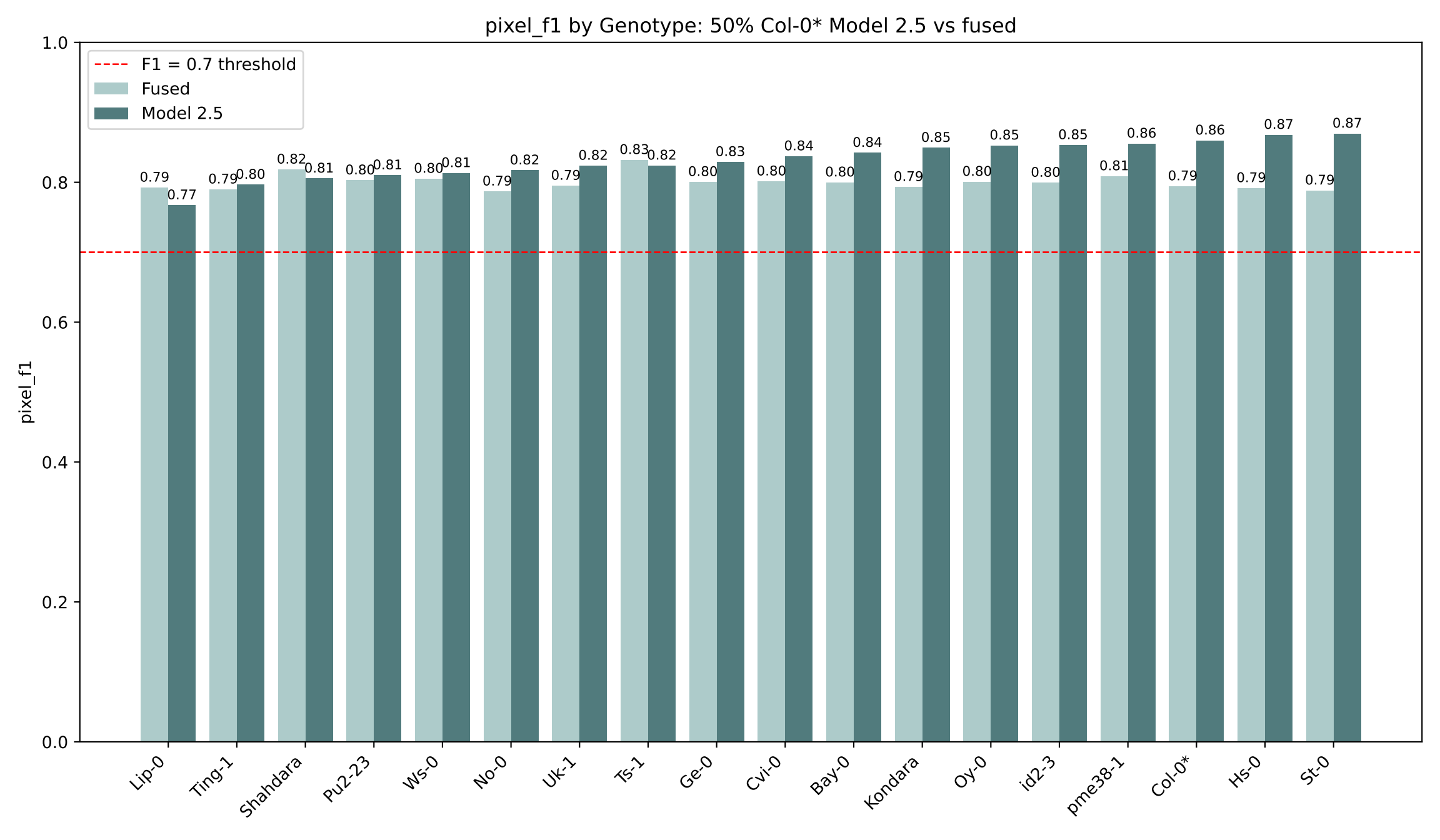
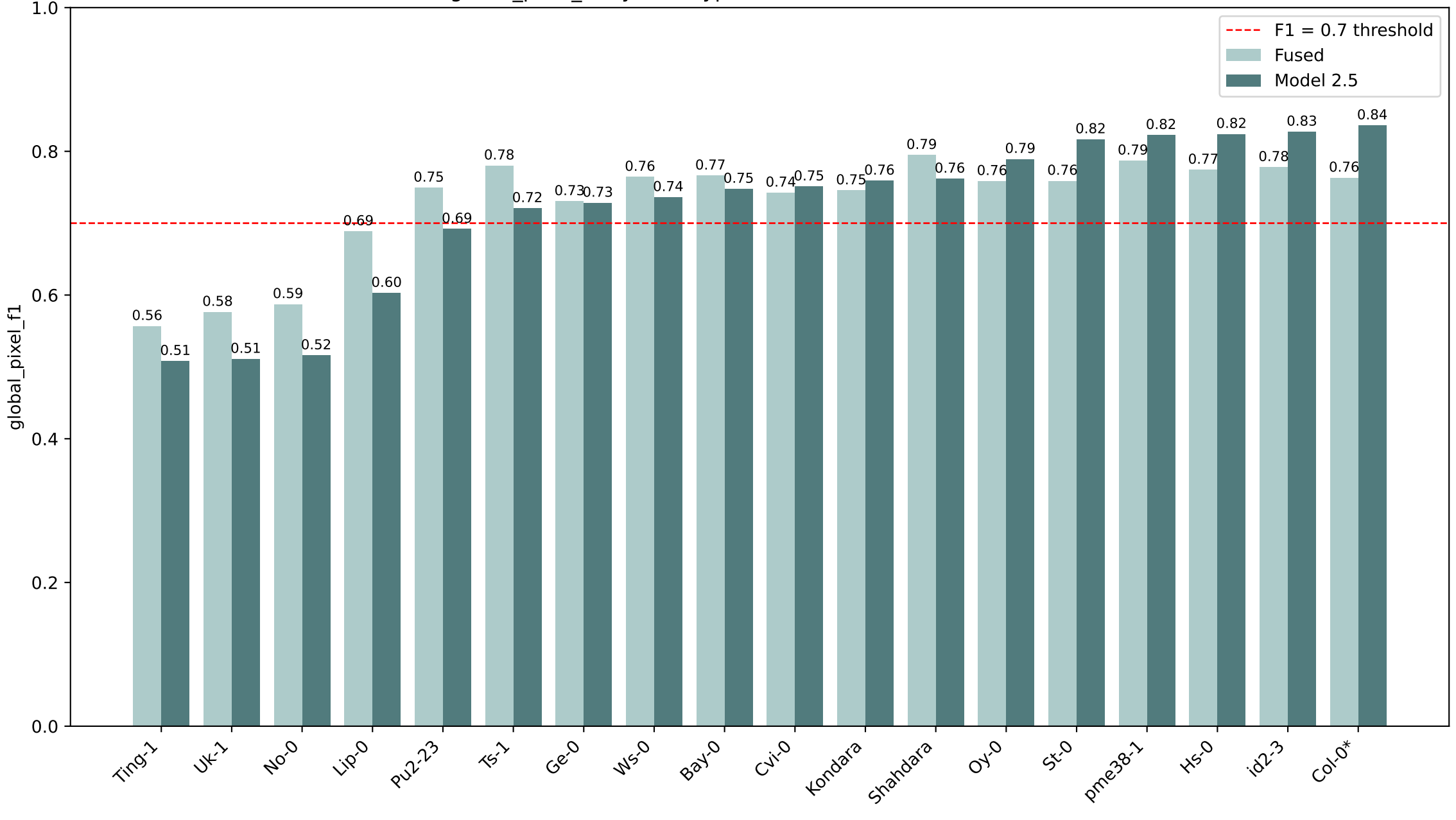

instance_f1 by Genotype: 100% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.44
0.42
0.43
0.43
0.53
0.54
0.54
0.55
0.62
0.65
0.61
0.68
0.65
0.69
0.70
0.70
0.69
0.72
0.74
0.76
0.75
0.76
0.71
0.78
0.76
0.79
0.75
0.79
0.77
0.80
0.79
0.81
0.77
0.82
0.82
0.84
Ting-1
No-0
Lip-0
Uk-1
Pu2-23
Ge-0
Cvi-0
Bay-0
Hs-0
Kondara
Ts-1
Ws-0
Col-0*
Oy-0
pme38-1
Shahdara
St-0
id2-3

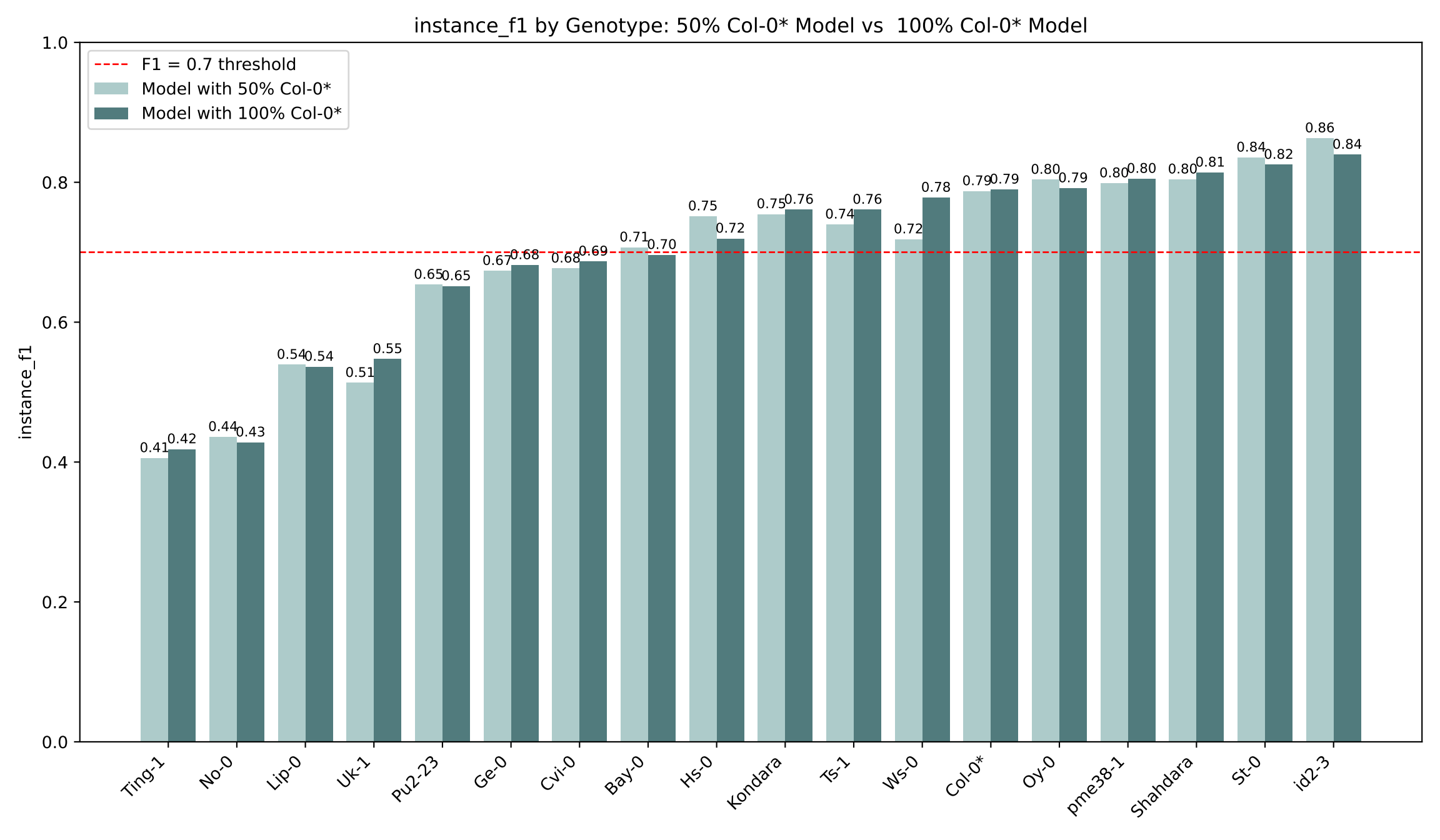

pixel_f1 by Genotype: 100% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.80 0.78
0.79 0.80
0.82 0.81
0.80 0.81
0.81 0.81
0.78 0.81
0.80 0.82
0.83 0.83
0.81 0.83
0.80 0.84
0.79 0.84
0.80 0.85
0.79 0.85
0.81 0.86
0.79 0.86
0.80 0.86
0.79 0.87
0.79 0.87
Lip-0
Ting-1
Shahdara
Pu2-23
Ws-0
No-0
Uk-1
Ts-1
Ge-0
Cvi-0
Kondara
Bay-0
Oy-0
pme38-1
Col-0*
id2-3
St-0
Hs-0

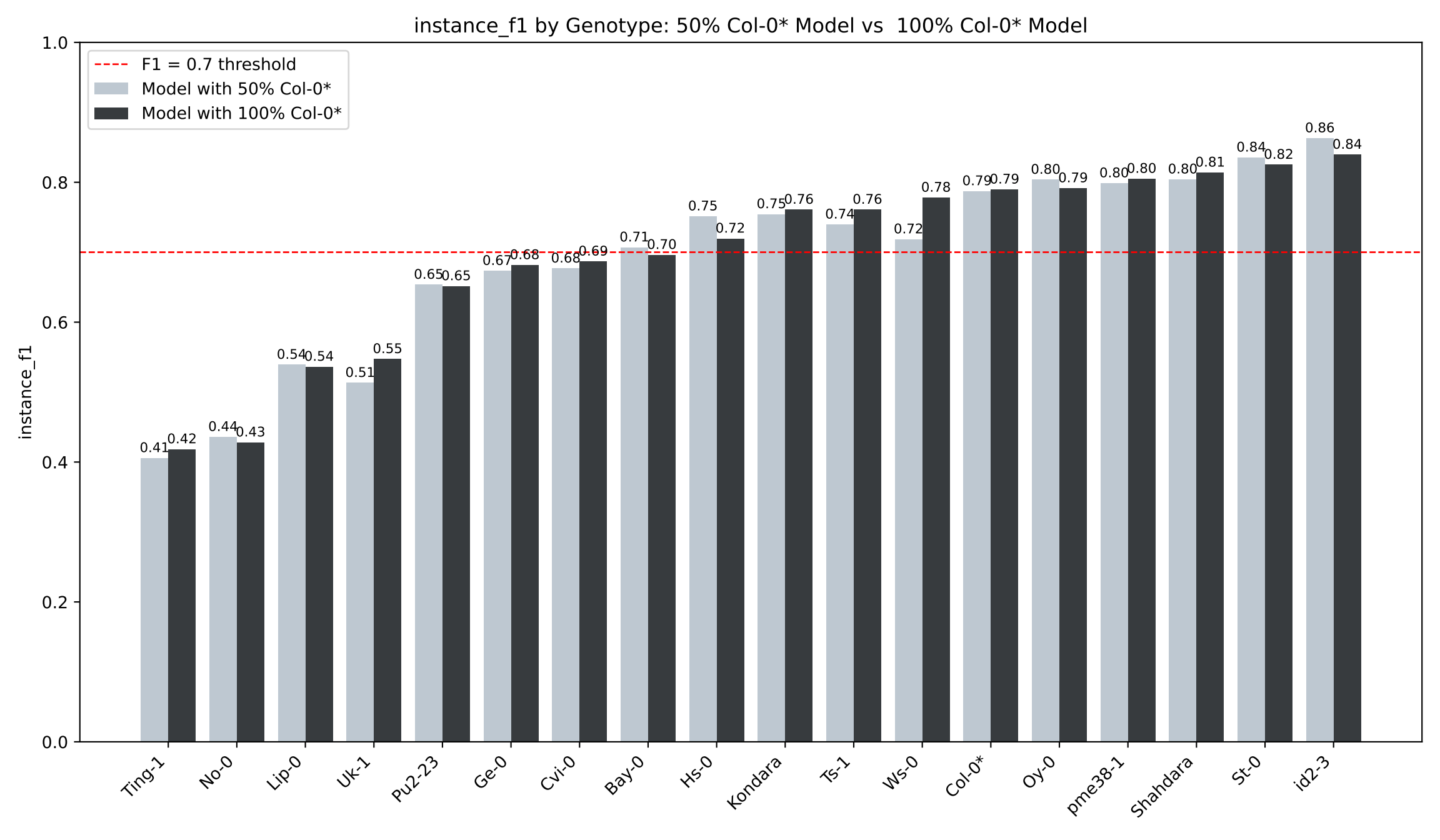

global_pixel_f1 by Genotype: 100% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.57
0.51
0.57
0.52
0.59
0.53
0.70
0.62
0.75
0.70
0.73
0.72
0.78
0.74
0.77
0.74
0.77
0.74
0.74
0.75
0.74
0.75
0.80
0.76
0.76
0.78
0.76
0.81
0.77
0.82
0.79
0.82
0.78
0.82
0.76
0.84
Uk-1
Ting-1
No-0
Lip-0
Pu2-23
Ge-0
Ts-1
Bay-0
Ws-0
Cvi-0
Kondara
Shahdara
Oy-0
St-0
Hs-0
pme38-1
id2-3
Col-0*

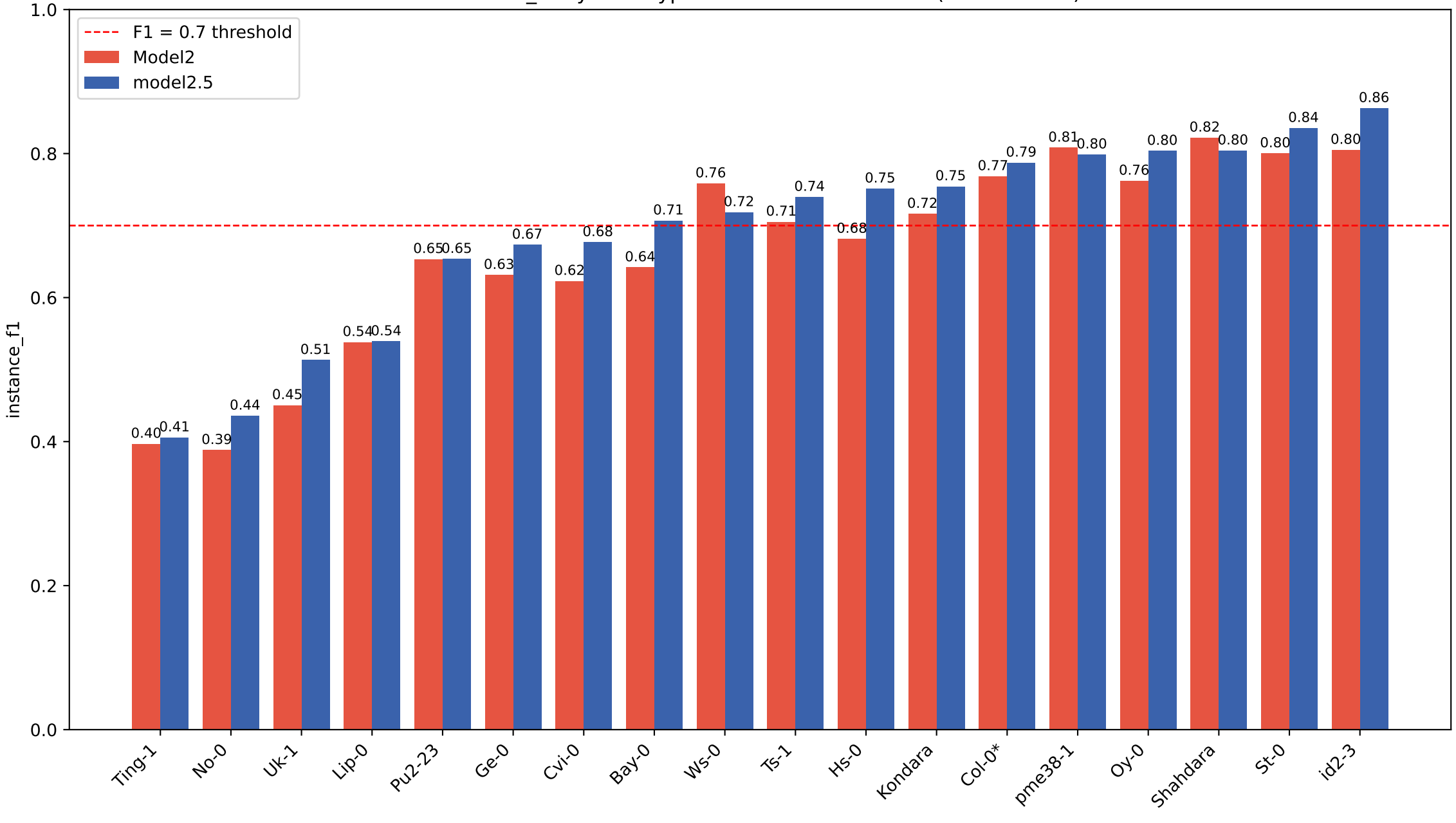

instance_f1 by Genotype: 50% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.44
0.41
0.44
0.44
0.52
0.51
0.53
0.54
0.62
0.65
0.62
0.67
0.64
0.68
0.72
0.71
0.67
0.72
0.73
0.74
0.73
0.75
0.74
0.75
0.76
0.79
0.76
0.80
0.74
0.80
0.78
0.80
0.79
0.84
0.83
0.86
Ting-1
No-0
Uk-1
Lip-0
Pu2-23
Ge-0
Cvi-0
Bay-0
Ws-0
Ts-1
Hs-0
Kondara
Col-0*
pme38-1
Oy-0
Shahdara
St-0
id2-3

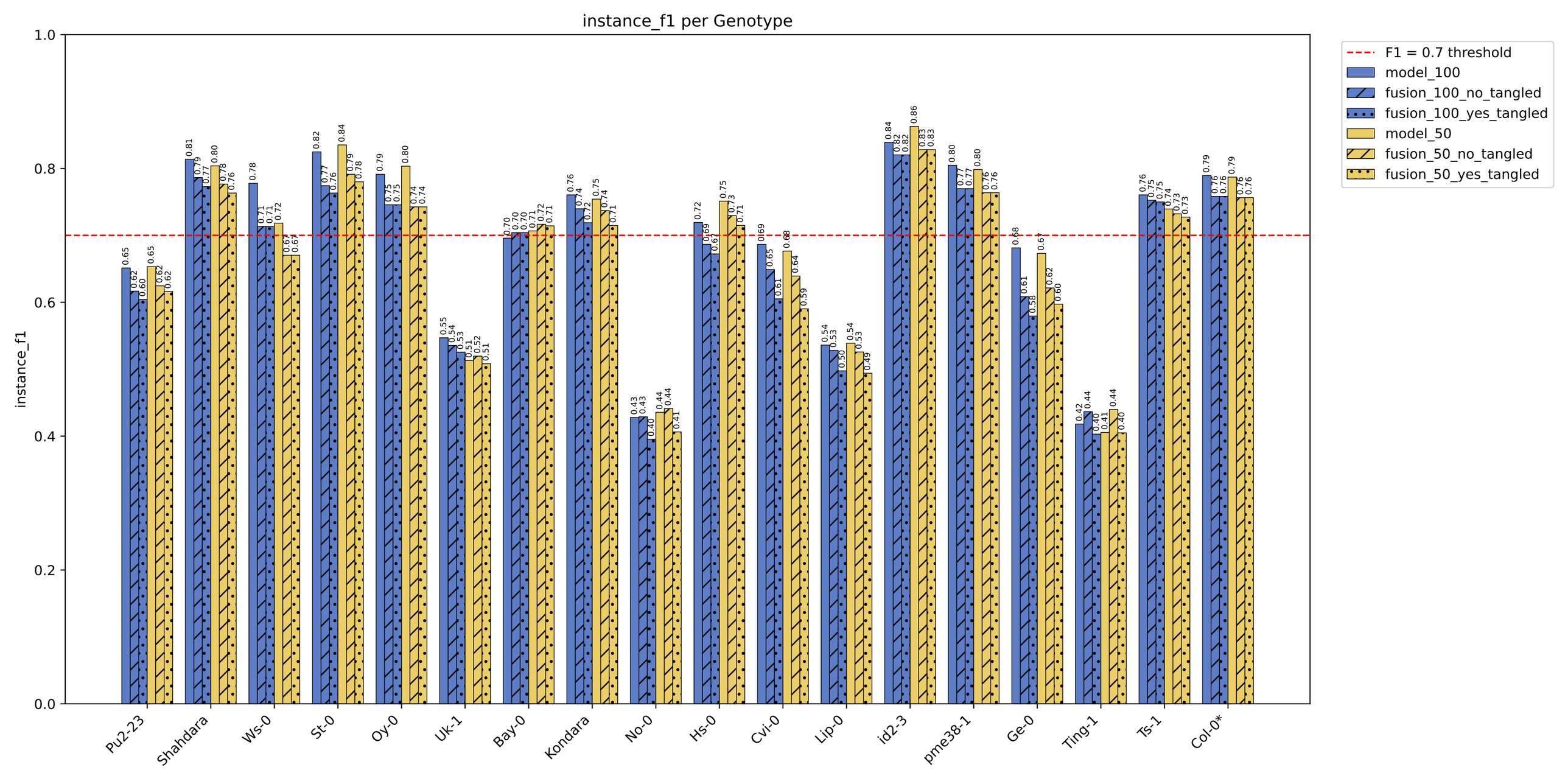

pixel_f1 by Genotype: 50% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.79 0.77
0.79 0.80
0.82 0.81
0.80 0.81
0.80 0.81
0.79 0.82
0.79 0.82
0.83 0.82
0.80 0.83
0.80 0.84
0.80 0.84
0.79 0.85
0.80 0.85
0.80 0.85
0.81 0.86
0.79 0.86
0.79 0.87
0.79 0.87
Lip-0
Ting-1
Shahdara
Pu2-23
Ws-0
No-0
Uk-1
Ts-1
Ge-0
Cvi-0
Bay-0
Kondara
Oy-0
id2-3
pme38-1
Col-0*
Hs-0
St-0

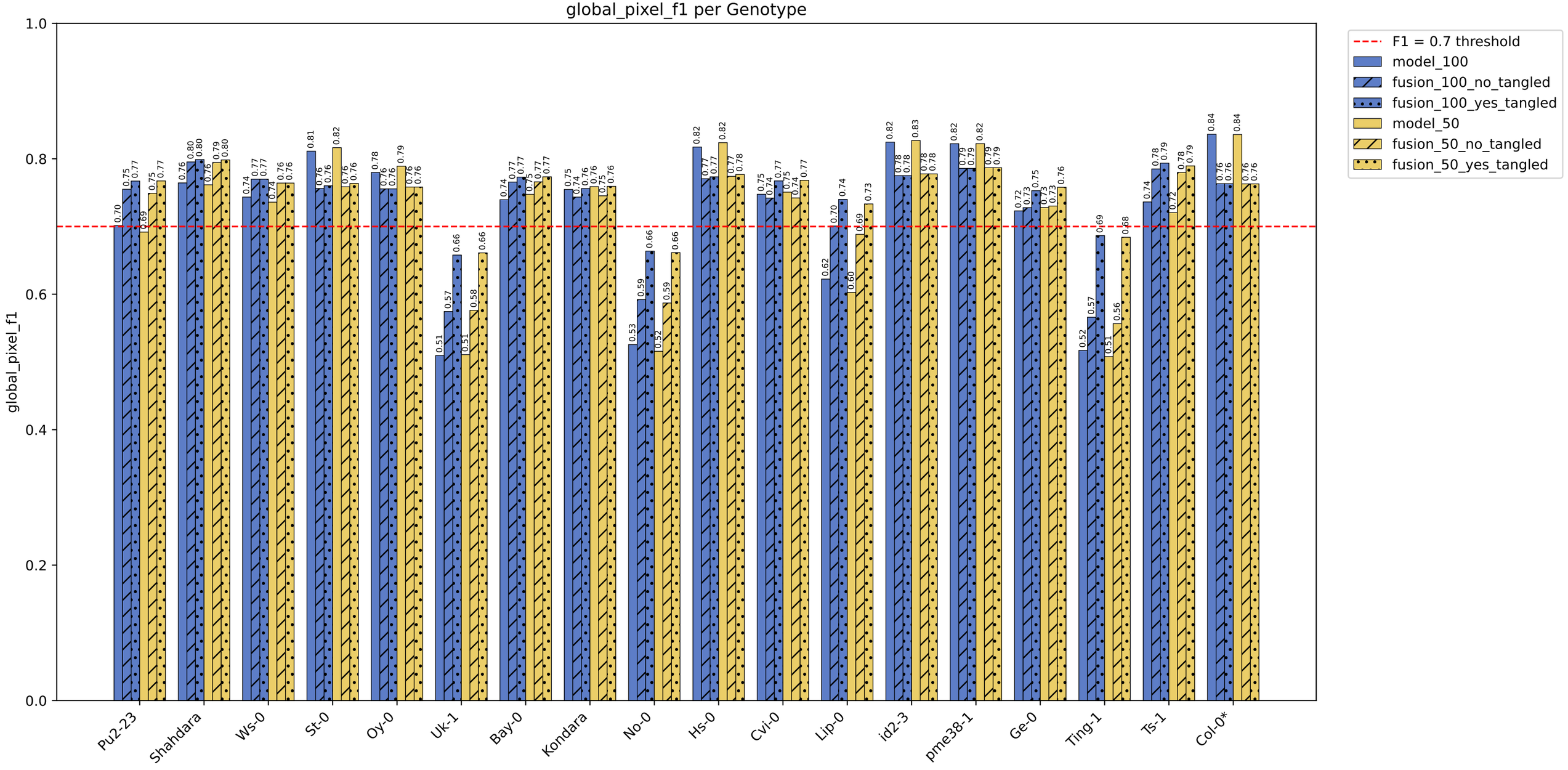

global_pixel_f1 by Genotype: 50% Col-0* Model 2.5 vs fused
F1 = 0.7 threshold
Fused
Model 2.5
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.56
0.51
0.58
0.51
0.59
0.52
0.69
0.60
0.75
0.69
0.78
0.72
0.73
0.73
0.76
0.74
0.77
0.75
0.74
0.75
0.75
0.76
0.79
0.76
0.76
0.79
0.76
0.82
0.79
0.82
0.77
0.82
0.78
0.83
0.76
0.84
Ting-1
Uk-1
No-0
Lip-0
Pu2-23
Ts-1
Ge-0
Ws-0
Bay-0
Cvi-0
Kondara
Shahdara
Oy-0
St-0
pme38-1
Hs-0
id2-3
Col-0*

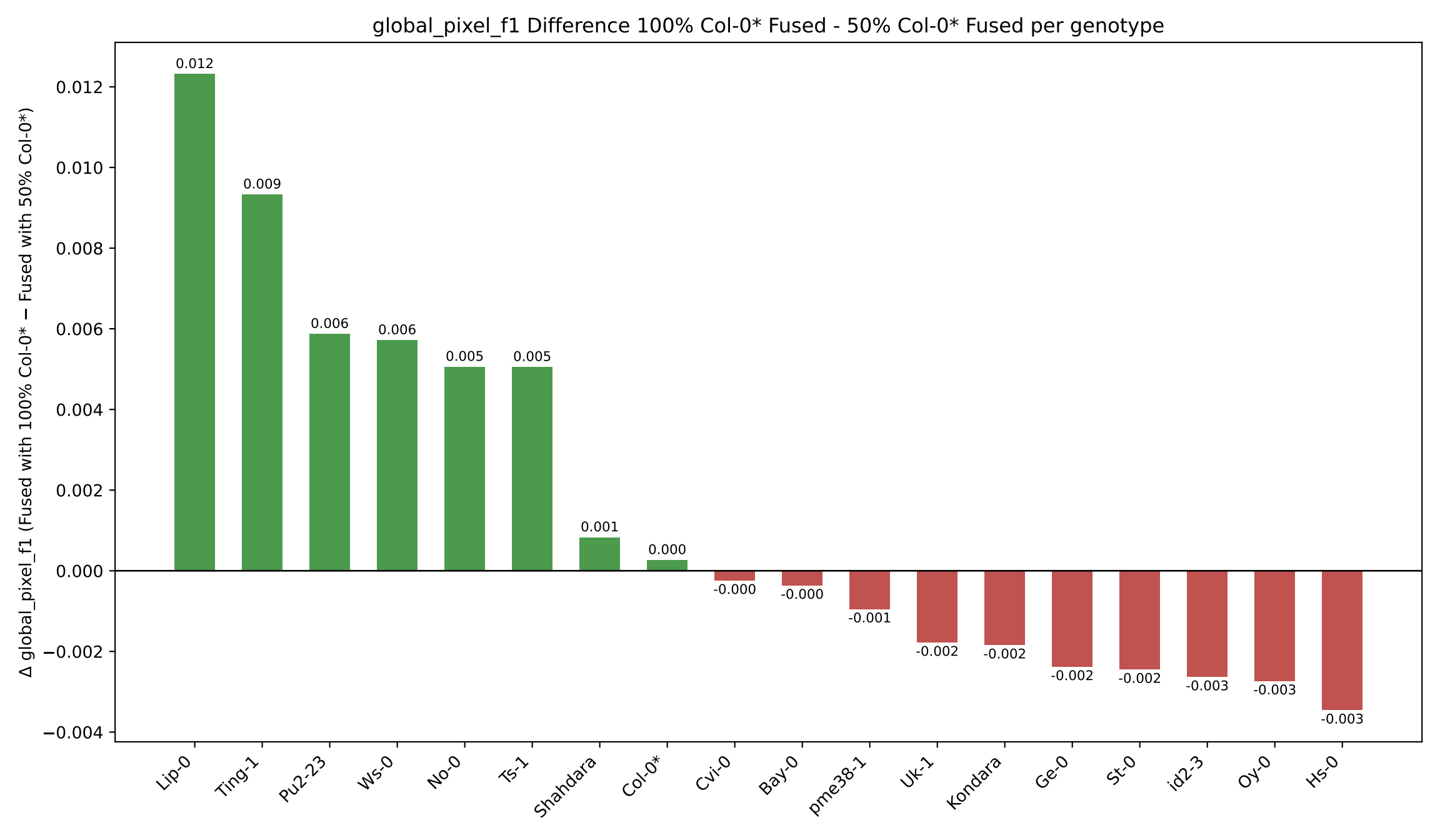

instance_f1 by Genotype: 50% Col-0* Model vs 100% Col-0* Model
F1 = 0.7 threshold
Model with 50% Col-0*
Model with 100% Col-0*
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.41 0.42
0.44 0.43
0.54 0.54
0.51 0.55
0.65 0.65
0.67 0.68
0.68 0.69
0.71 0.70
0.75 0.72
0.75 0.76
0.74 0.76
0.72 0.78
0.79 0.79
0.80 0.79
0.80 0.80
0.80 0.81
0.84 0.82
0.86 0.84
Ting-1
No-0
Lip-0
Uk-1
Pu2-23
Ge-0
Cvi-0
Bay-0
Hs-0
Kondara
Ts-1
Ws-0
Col-0*
Oy-0
pme38-1
Shahdara
St-0
id2-3

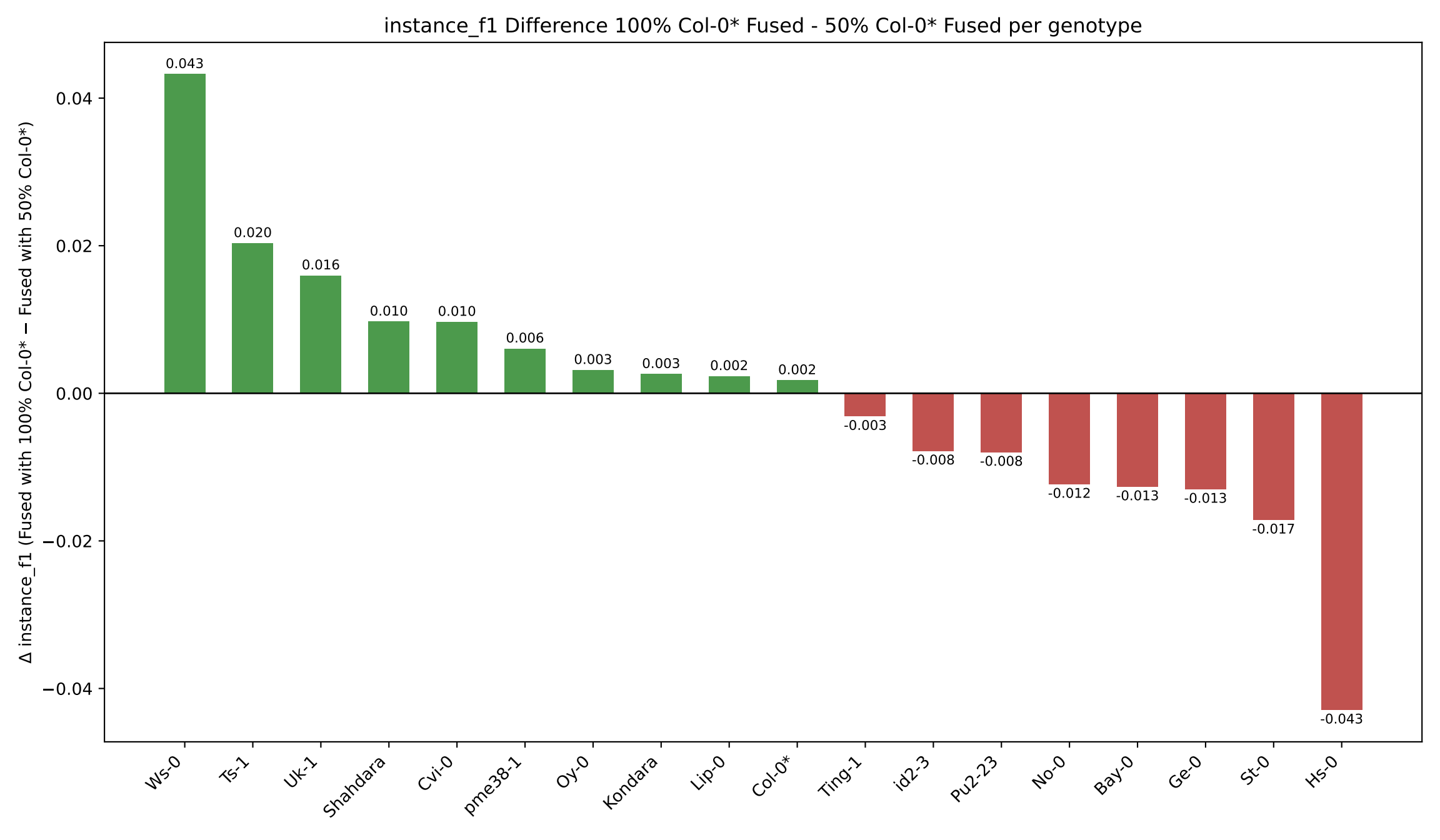

instance_f1 by Genotype: 50% Col-0* Model vs 100% Col-0* Model
F1 = 0.7 threshold
Model with 50% Col-0*
Model with 100% Col-0*
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.41 0.42
0.44 0.43
0.54 0.54
0.51 0.55
0.65 0.65
0.67 0.68
0.68 0.69
0.71 0.70
0.75 0.72
0.75 0.76
0.74 0.76
0.72 0.78
0.79 0.79
0.80 0.79
0.80 0.80
0.80 0.81
0.84 0.82
0.86 0.84
Ting-1
No-0
Lip-0
Uk-1
Pu2-23
Ge-0
Cvi-0
Bay-0
Hs-0
Kondara
Ts-1
Ws-0
Col-0*
Oy-0
pme38-1
Shahdara
St-0
id2-3


instance_f1 by Genotype: Model 2 Vs Model 2.5 (100% Col-0*)
F1 = 0.7 threshold
Model2
model2.5
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.40 0.41
0.39 0.44
0.45 0.51
0.54 0.54
0.65 0.65
0.63 0.67
0.62 0.68
0.64 0.71
0.76 0.72
0.71 0.74
0.68 0.75
0.72 0.75
0.77 0.79
0.81 0.80
0.76 0.80
0.82 0.80
0.80 0.84
0.80 0.86
Ting-1
No-0
Uk-1
Lip-0
Pu2-23
Ge-0
Cvi-0
Bay-0
Ws-0
Ts-1
Hs-0
Kondara
Col-0*
pme38-1
Oy-0
Shahdara
St-0
id2-3


instance_f1 per Genotype
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
model_100
fusion_100_no_tangled
fusion_100_yes_tangled
model_50
fusion_50_no_tangled
fusion_50_yes_tangled
Pu2-23
Shahdara
Ws-0
St-0
Oy-0
Uk-1
Bay-0
Kondara
No-0
Hs-0
Cvi-0
Lip-0
id2-3
pme38-1
Ge-0
Ting-1
Ts-1
Col-0*


global_pixel_f1 per Genotype
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
model_100
fusion_100_no_tangled
fusion_100_yes_tangled
model_50
fusion_50_no_tangled
fusion_50_yes_tangled
Pu2-23
Shahdara
Ws-0
St-0
Oy-0
Uk-1
Bay-0
Kondara
No-0
Hs-0
Cvi-0
Lip-0
id2-3
pme38-1
Ge-0
Ting-1
Ts-1
Col-0*

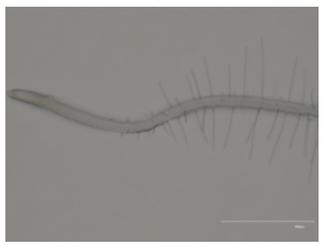

global_pixel_f1 Difference 100% Col-0* Fused - 50% Col-0* Fused per genotype
Δ global_pixel_f1 (Fused with 100% Col-0* − Fused with 50% Col-0*)
0.012
0.010
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
0.012
0.009
0.006
0.006
0.005
0.005
0.001
0.000
-0.000
-0.000
-0.001
-0.002
-0.002
-0.002
-0.002
-0.003
-0.003
-0.003
Lip-0
Ting-1
Pu2-23
Ws-0
No-0
Ts-1
Shahdara
Col-0*
Cvi-0
Bay-0
pme38-1
Uk-1
Kondara
Ge-0
St-0
id2-3
Oy-0
Hs-0


instance_f1 Difference 100% Col-0* Fused - 50% Col-0* Fused per genotype
Δ instance_f1 (Fused with 100% Col-0* − Fused with 50% Col-0*)
0.04
0.02
0.00
−0.02
−0.04
0.043
0.020
0.016
0.010
0.010
0.006
0.003
0.003
0.002
0.002
-0.003
-0.008
-0.008
-0.012
-0.013
-0.013
-0.017
-0.043
Ws-0
Ts-1
Uk-1
Shahdara
Cvi-0
pme38-1
Oy-0
Kondara
Lip-0
Col-0*
Ting-1
id2-3
Pu2-23
No-0
Bay-0
Ge-0
St-0
Hs-0

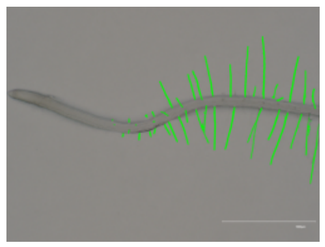
Original vs. Fused (50%, wo) Prediction of Good Ecotypes

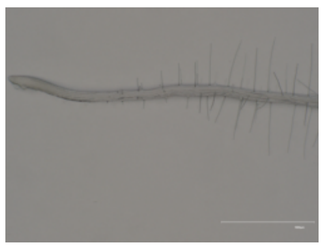
St-0

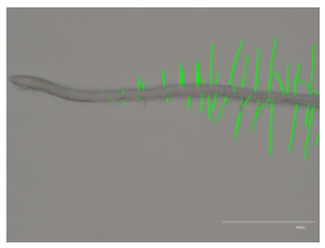
Original



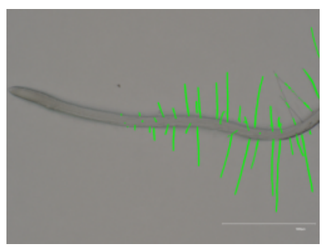
Prediction





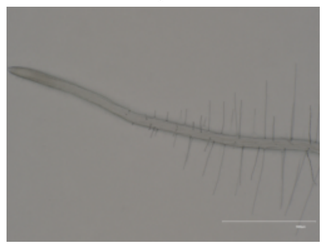




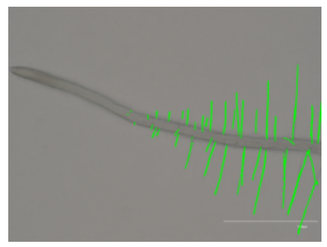


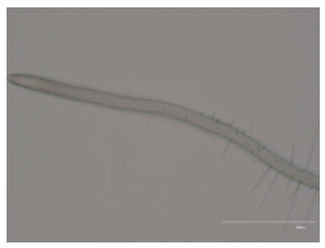
Hs-0

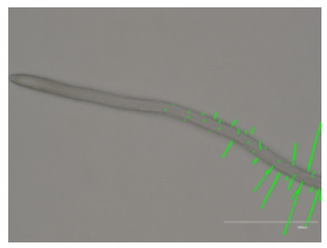
Original

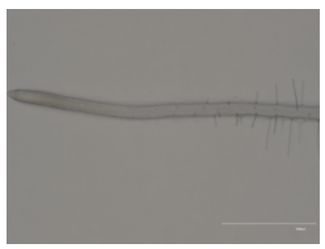


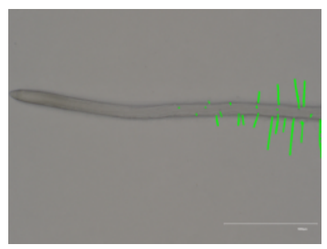
Prediction





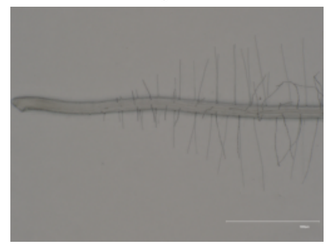




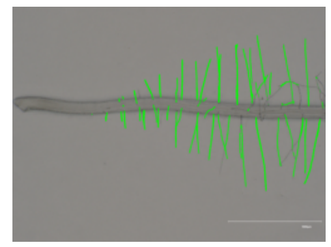


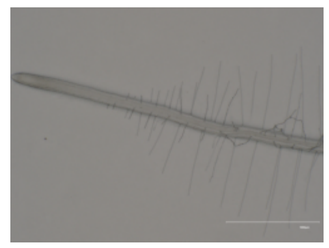
Cvi-0

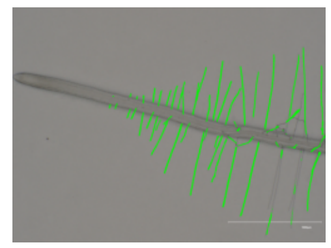
Original

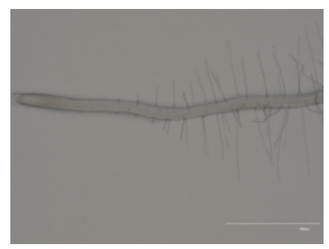


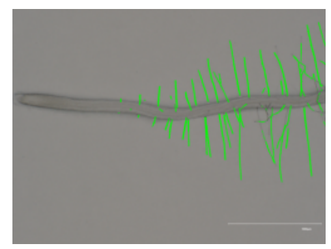
Prediction









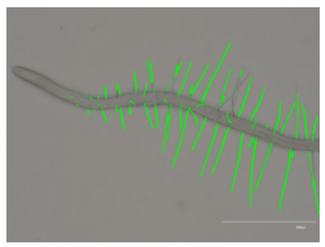


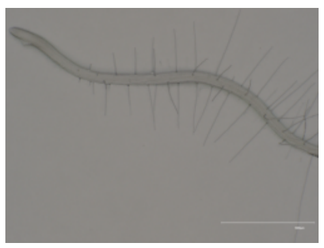
Kondara

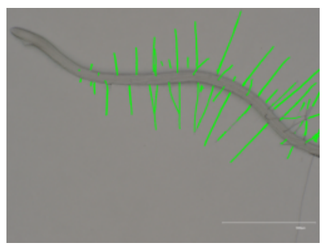
Original

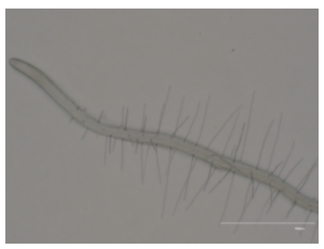


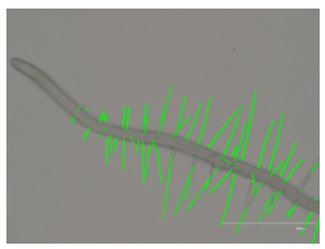
Prediction





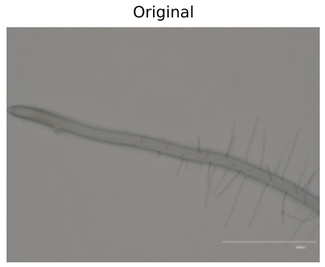




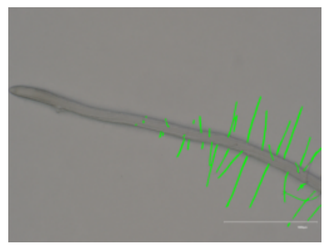


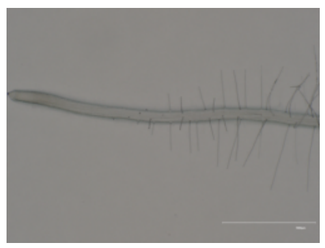
Oy-0

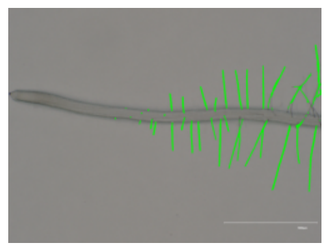
Original

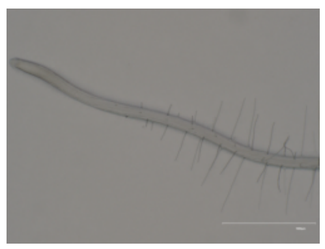


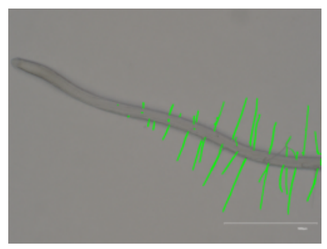
Prediction





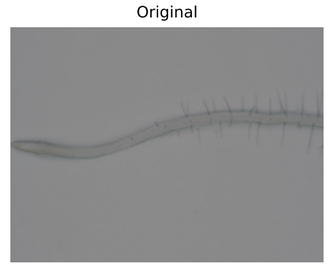




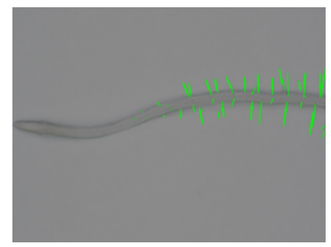


pme38-1

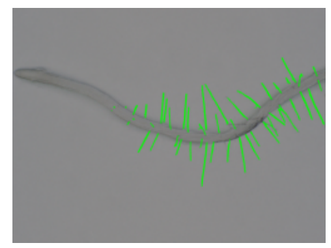
Original



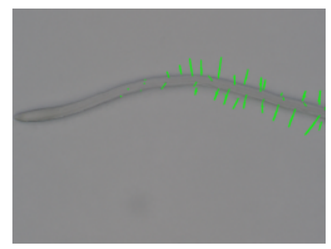
Prediction







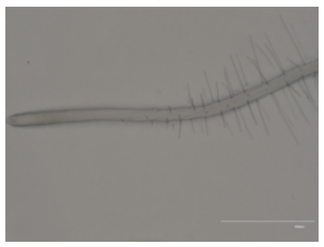




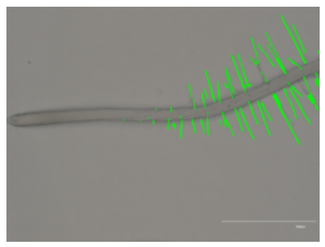
Original vs. Fused (50%, wo) Prediction of Bad Ecotypes

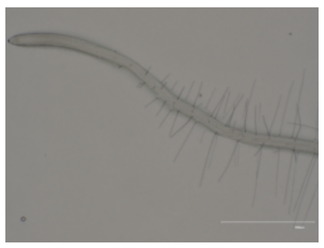
Pu2-23

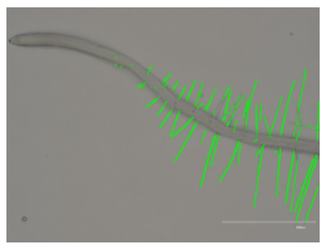
Original

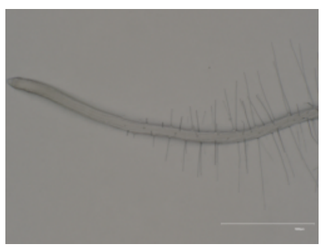


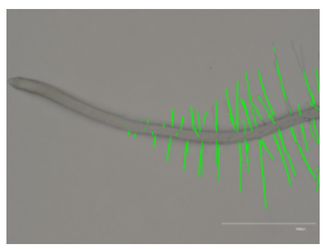
Prediction





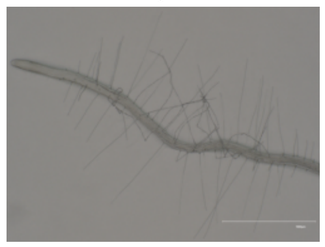




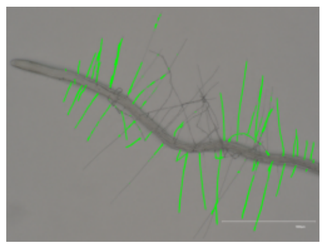


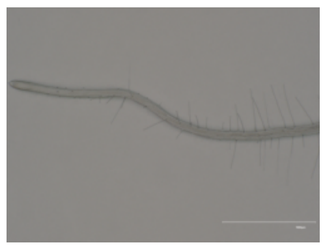
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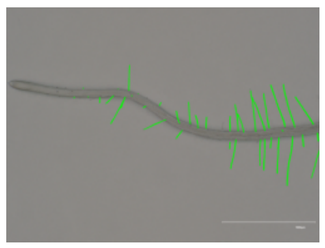
Original

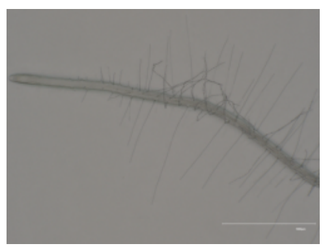


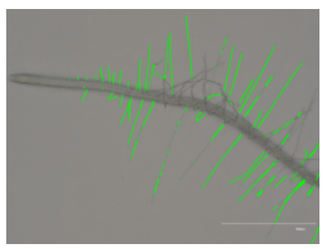
Prediction





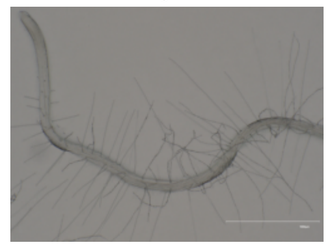




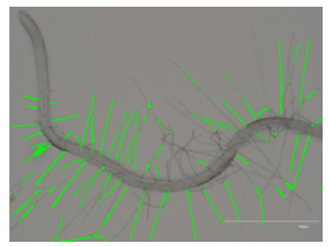


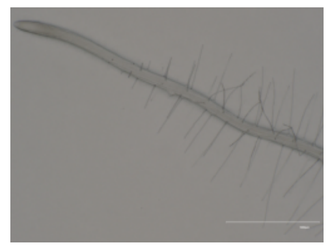
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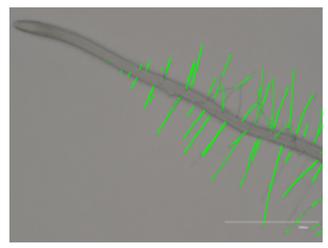
Original

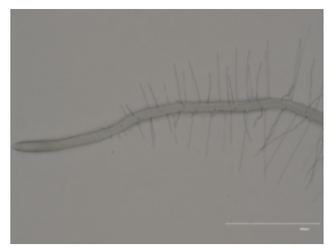


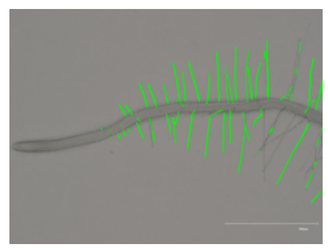
Prediction





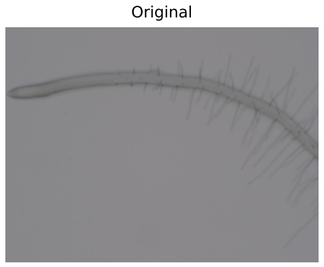




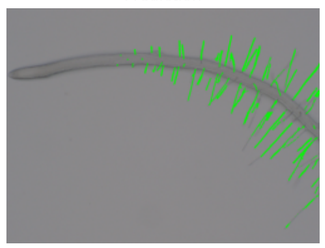


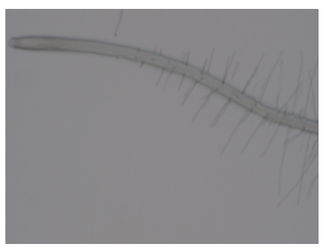
Lip-0

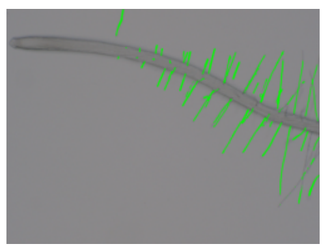
Original

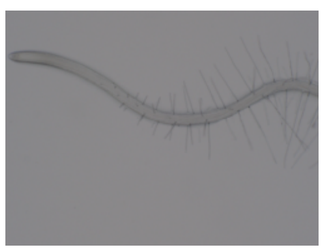


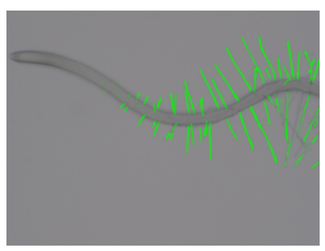
Prediction





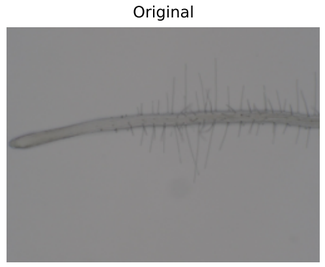




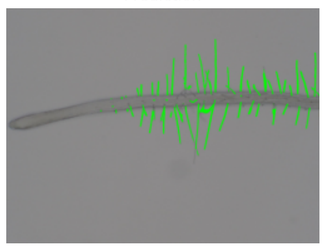


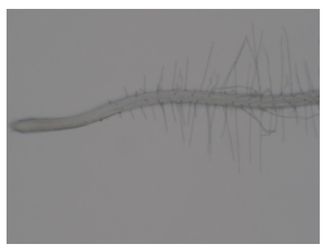
Ge-0

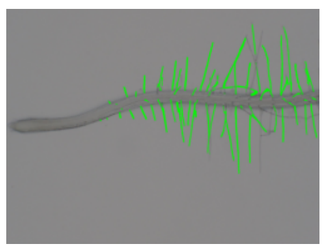
Original



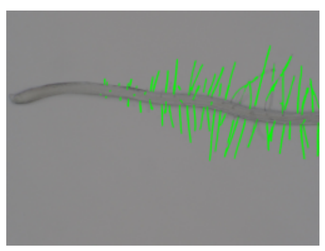
Prediction





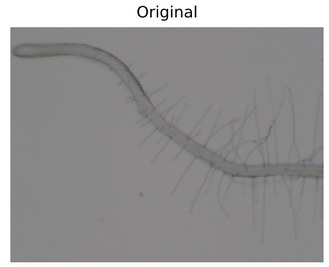




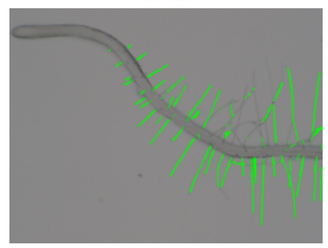


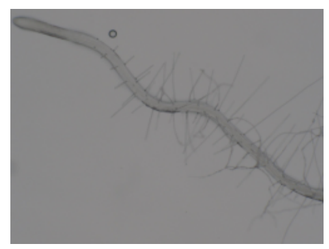
Ting-1

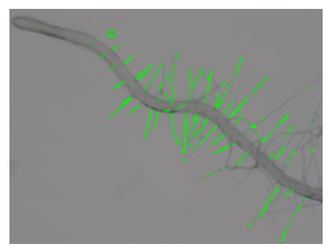
Original

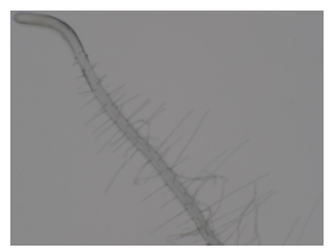


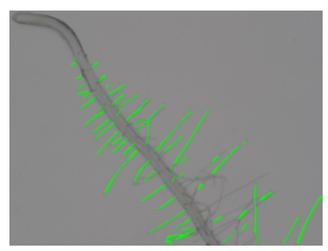
Prediction

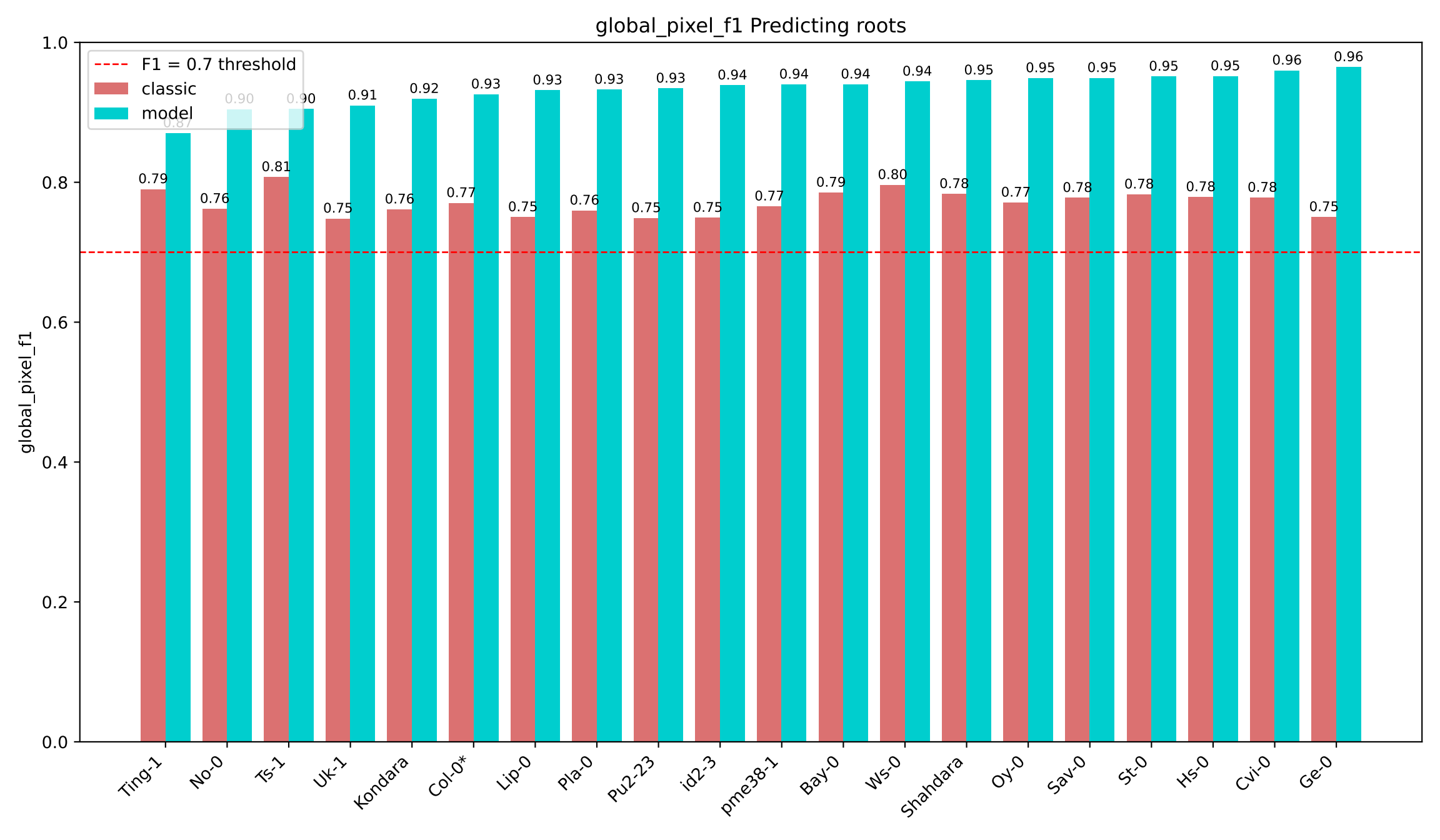






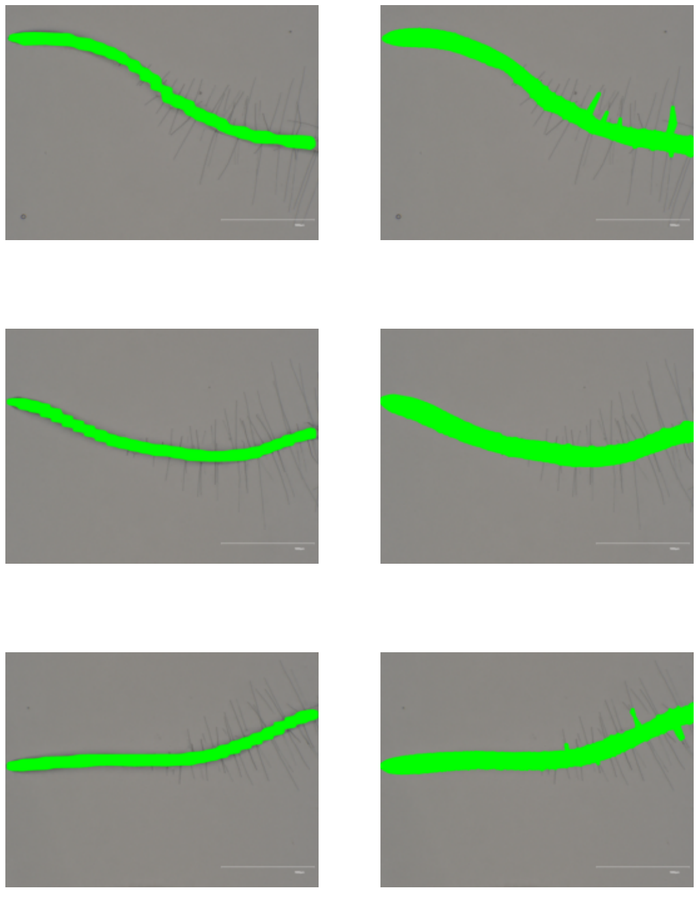




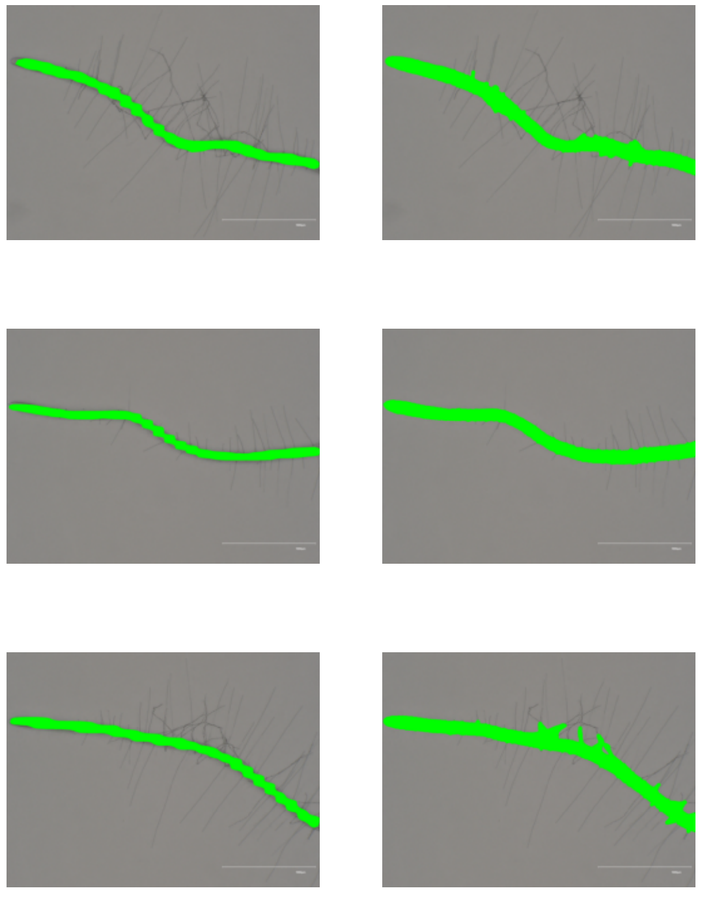

global_pixel_f1 Predicting roots
F1 = 0.7 threshold
classic
model
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.79
0.87
0.76
0.90
0.81
0.90
0.75
0.91
0.76
0.92
0.77
0.93
0.75
0.93
0.76
0.93
0.75
0.93
0.75
0.94
0.77
0.94
0.79
0.94
0.80
0.94
0.78
0.95
0.77
0.95
0.78
0.95
0.78
0.95
0.78
0.95
0.75
0.96
0.96
Ting-1
No-0
Ts-1
Uk-1
Kondara
Col-0*
Lip-0
Pla-0
Pu2-23
id2-3
pme38-1
Bay-0
Ws-0
Shahdara
Oy-0
Sav-0
St-0
Hs-0
Cvi-0
Ge-0

Model Root Prediction vs.Classic Root Prediction of Bad Ecotypes

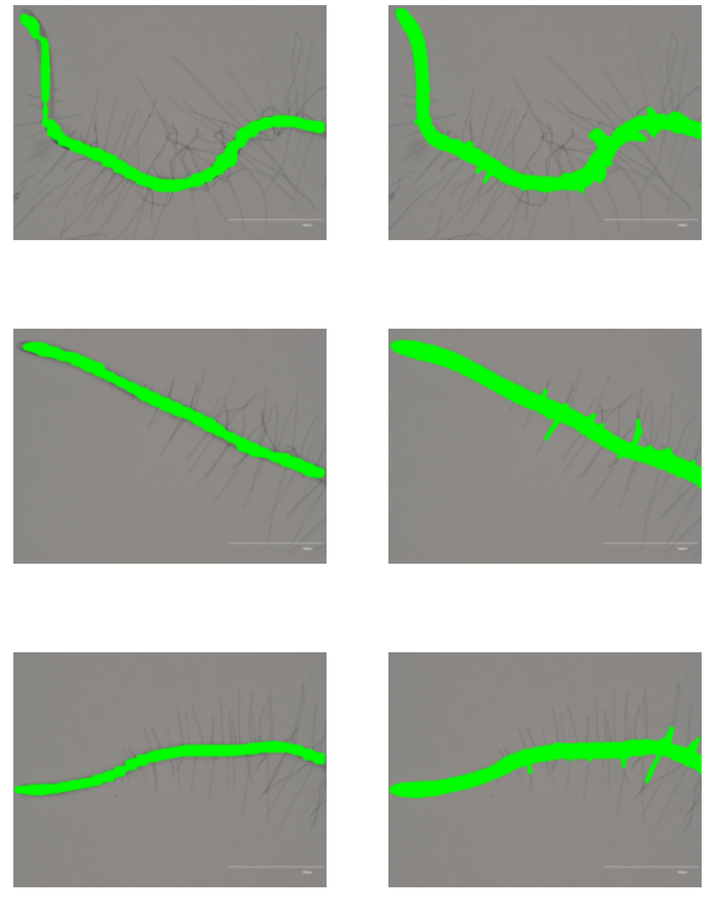
Pu2-23



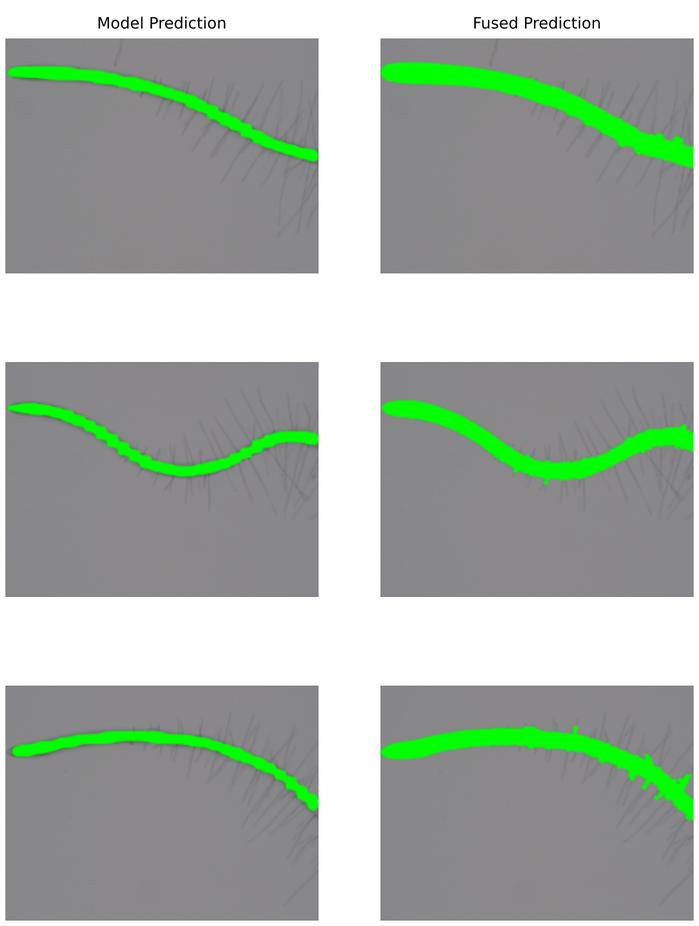
Uk-1



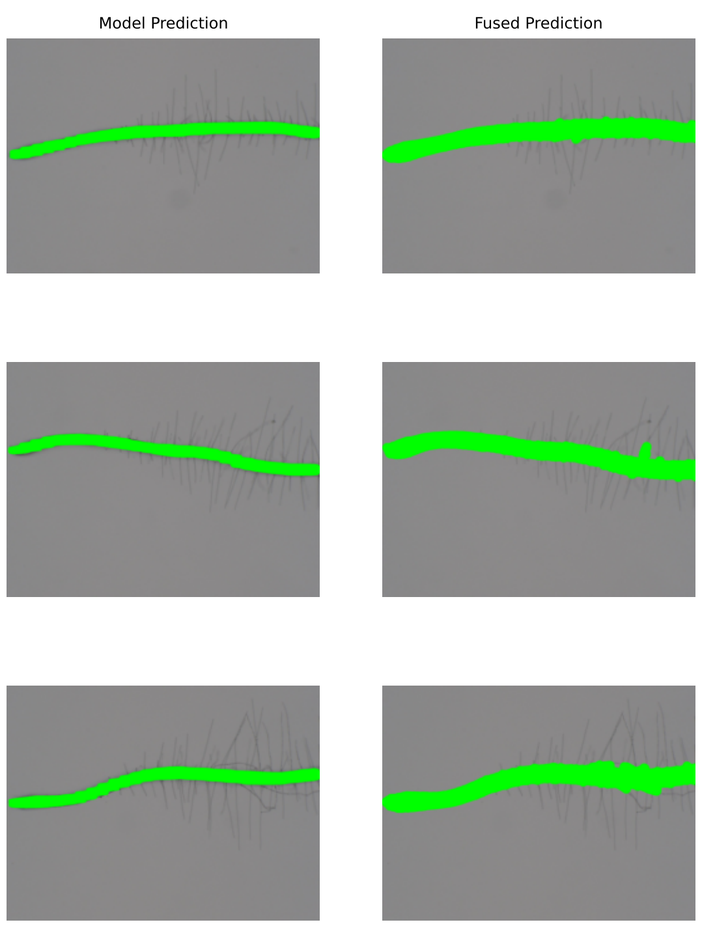
No-0



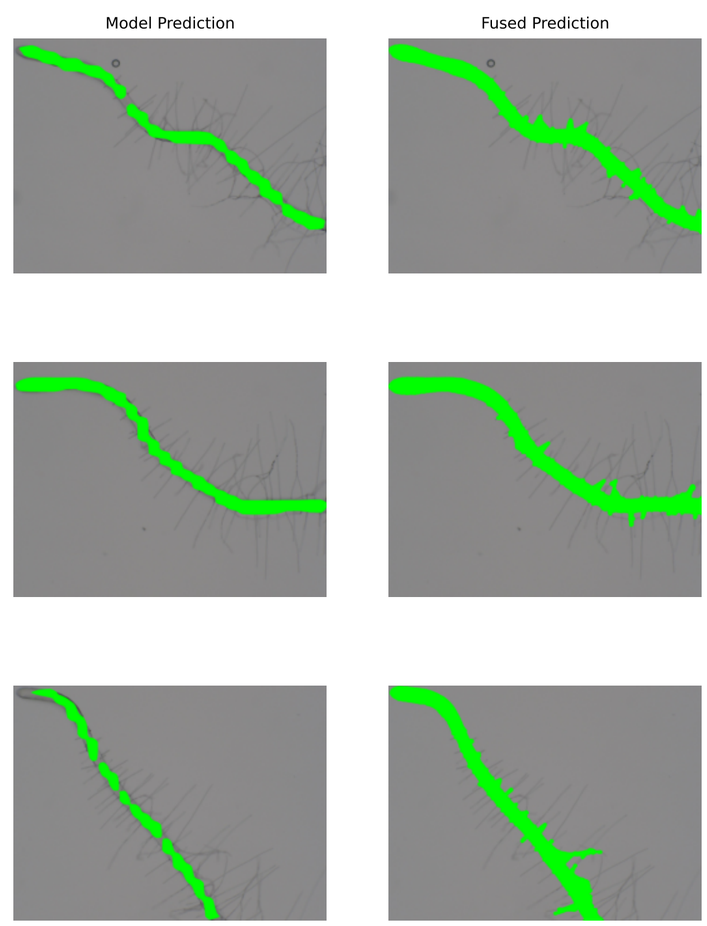
Lip-0



Ge-0



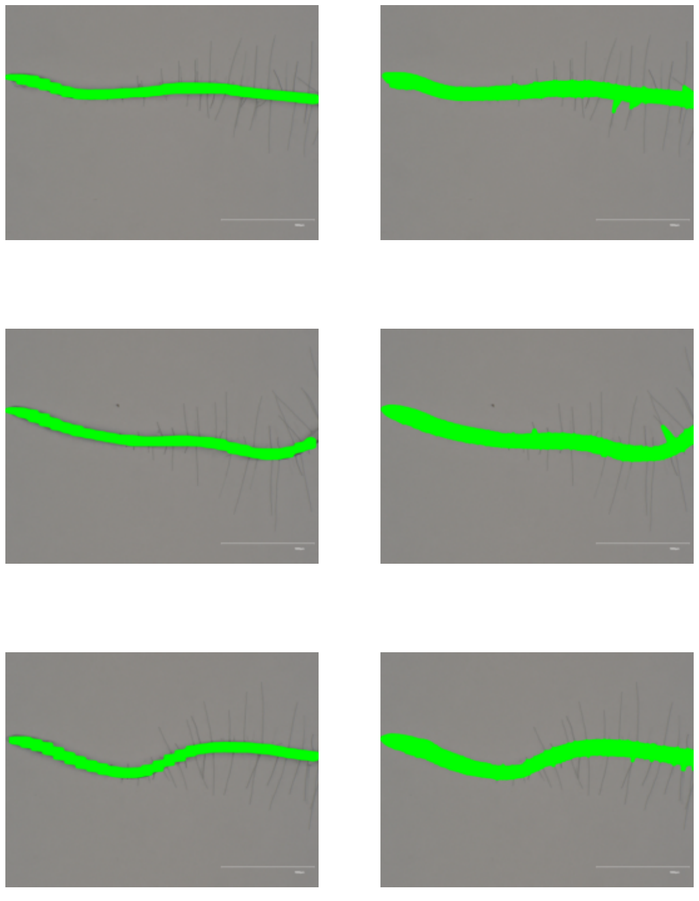
Ting-1



Model Root Prediction vs.Classic Root Prediction of Good Ecotypes

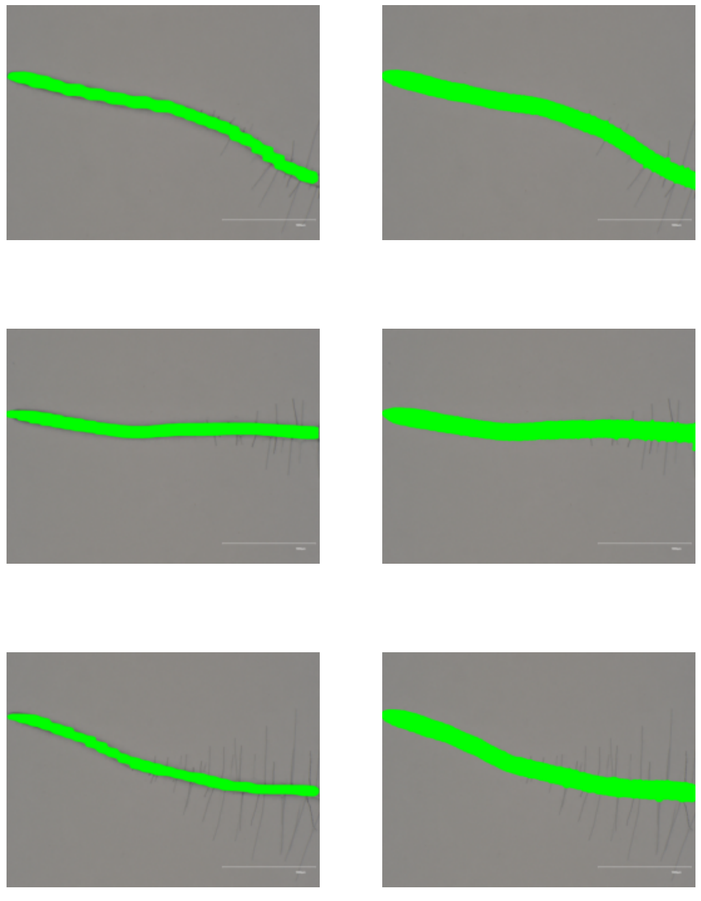
St-0

Model Prediction | Fused Prediction



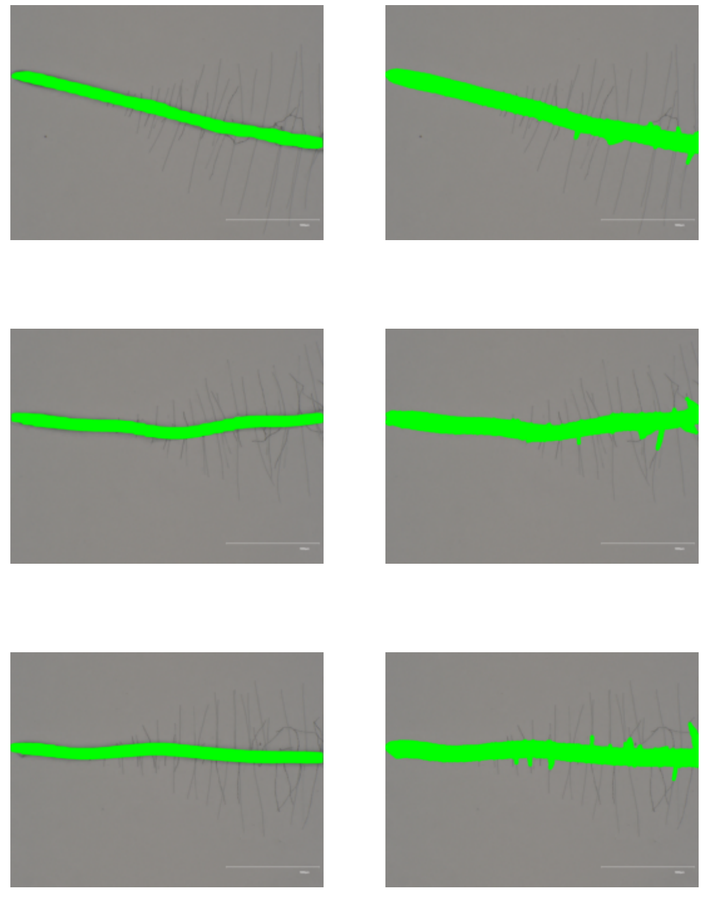
Hs-0

Model Prediction | Fused Prediction



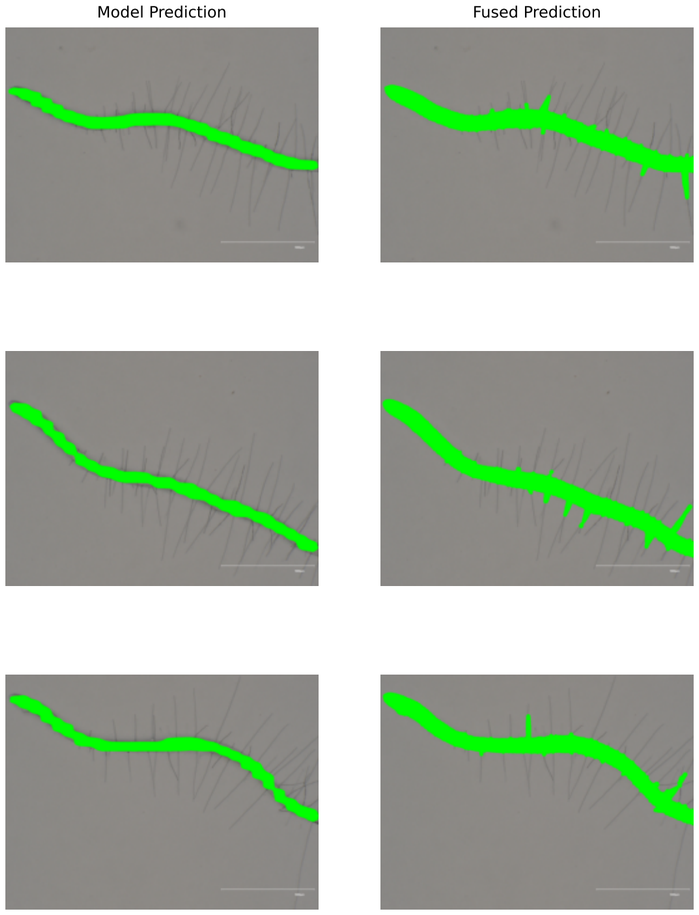
Cvi-0

Model Prediction | Fused Prediction



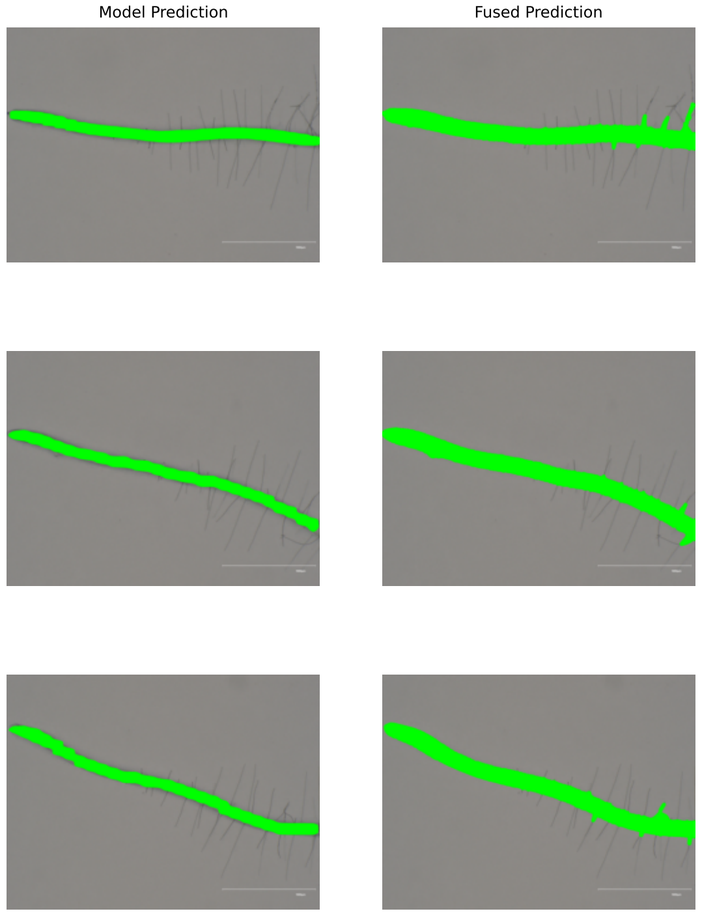
Kondara

Model Prediction | Fused Prediction



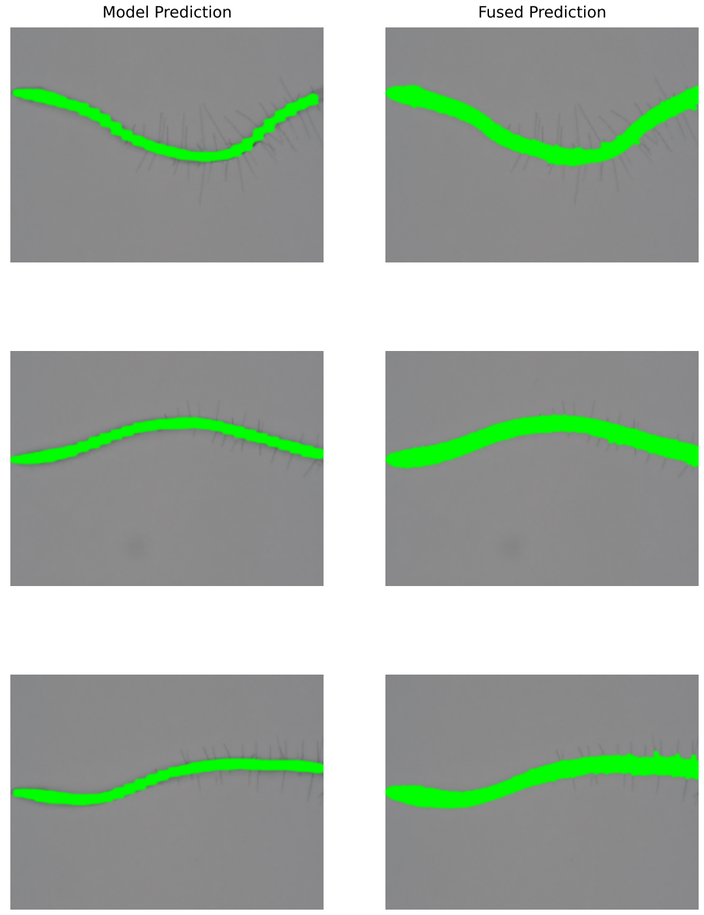
Oy-0

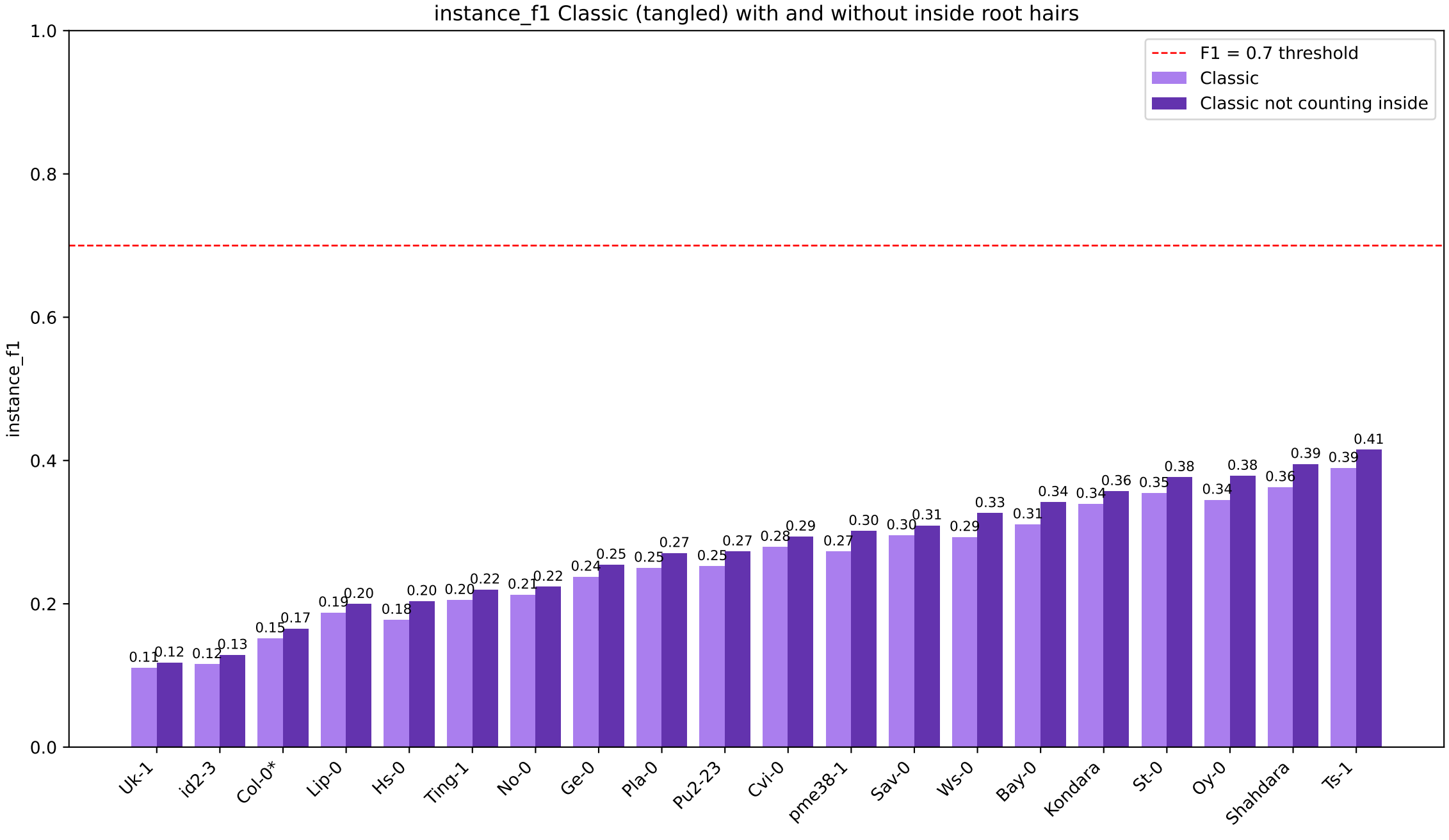
Model Prediction | Fused Prediction

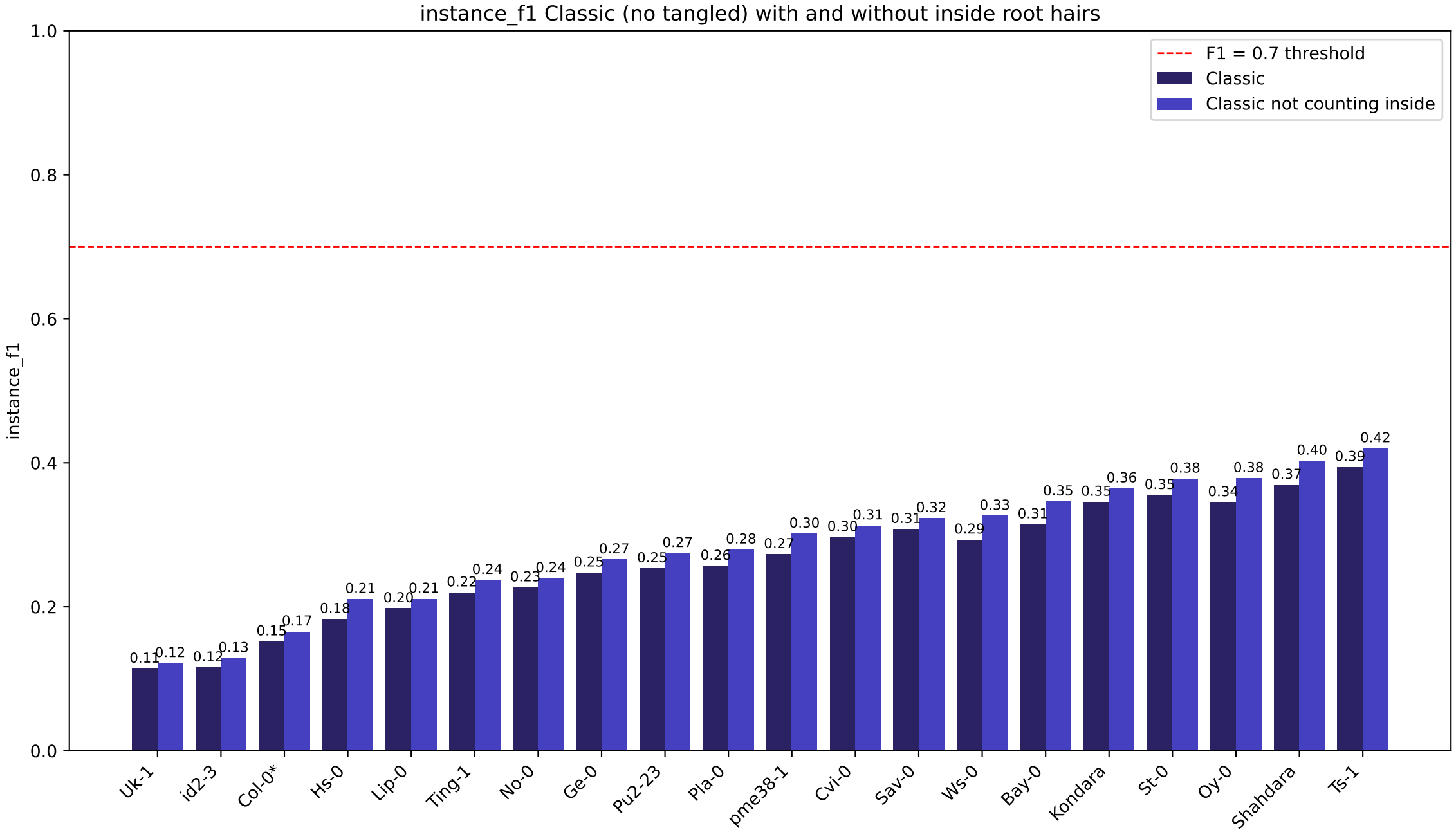


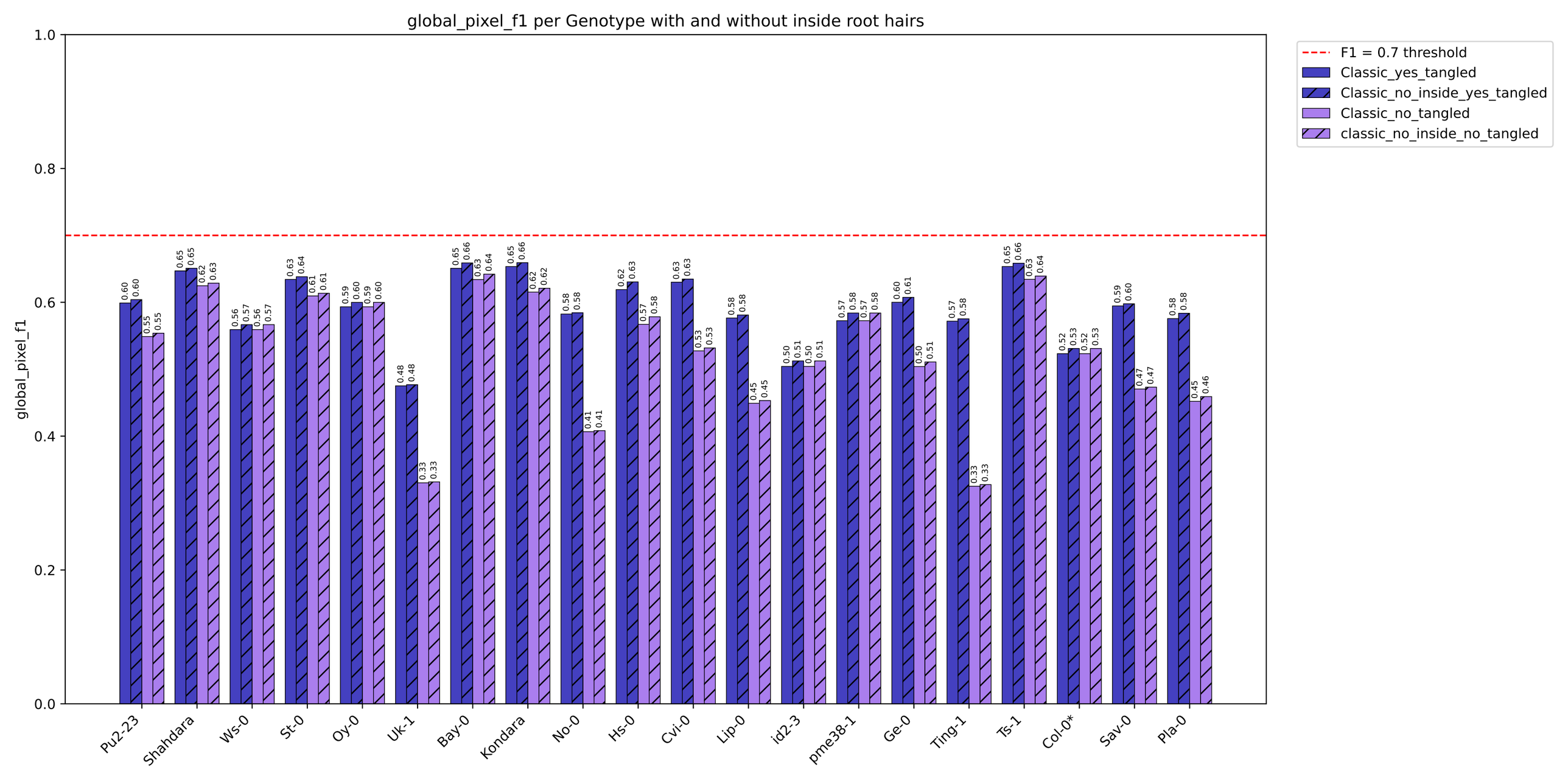
pme38-1

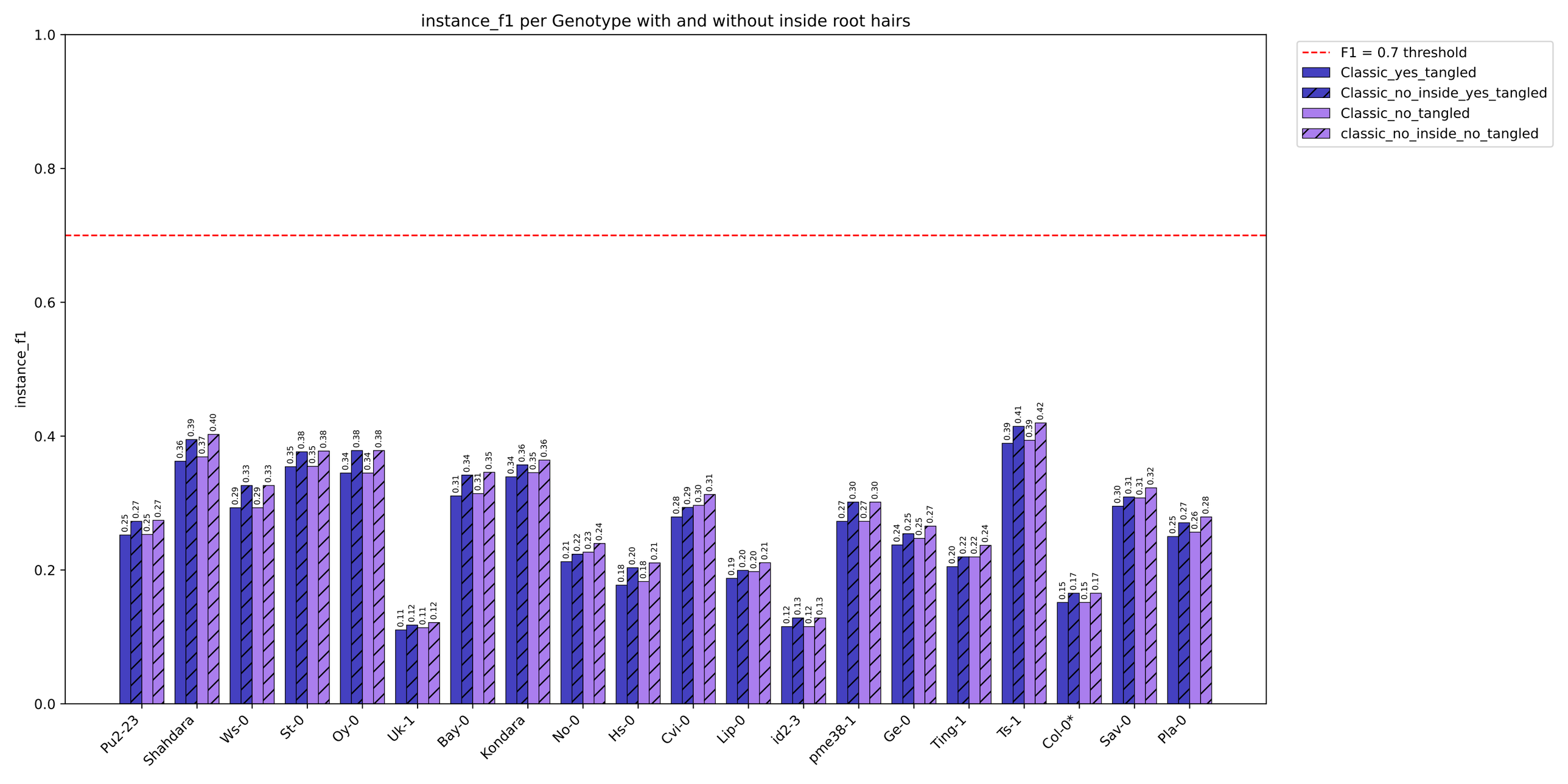
Model Prediction | Fused Prediction

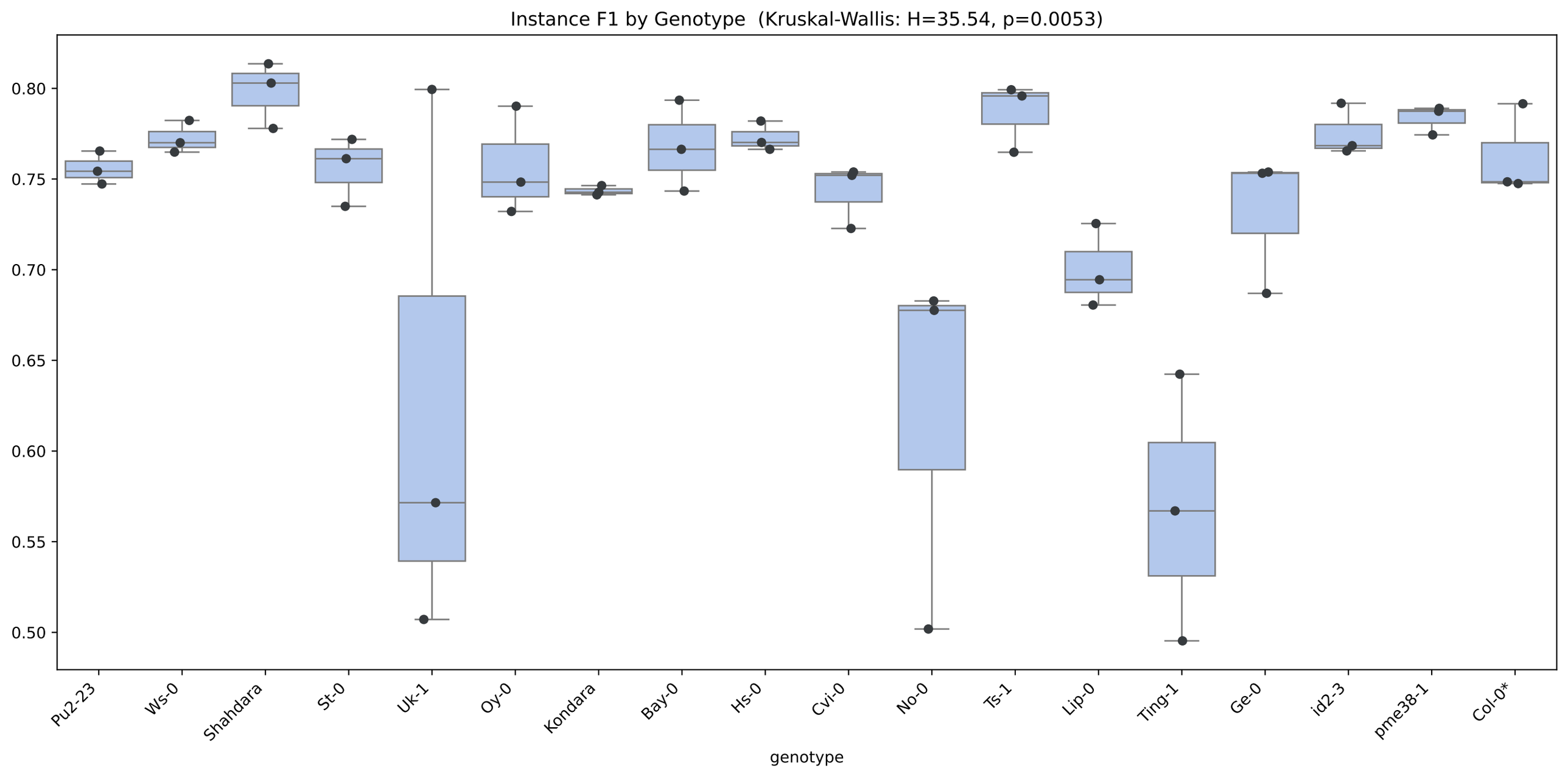


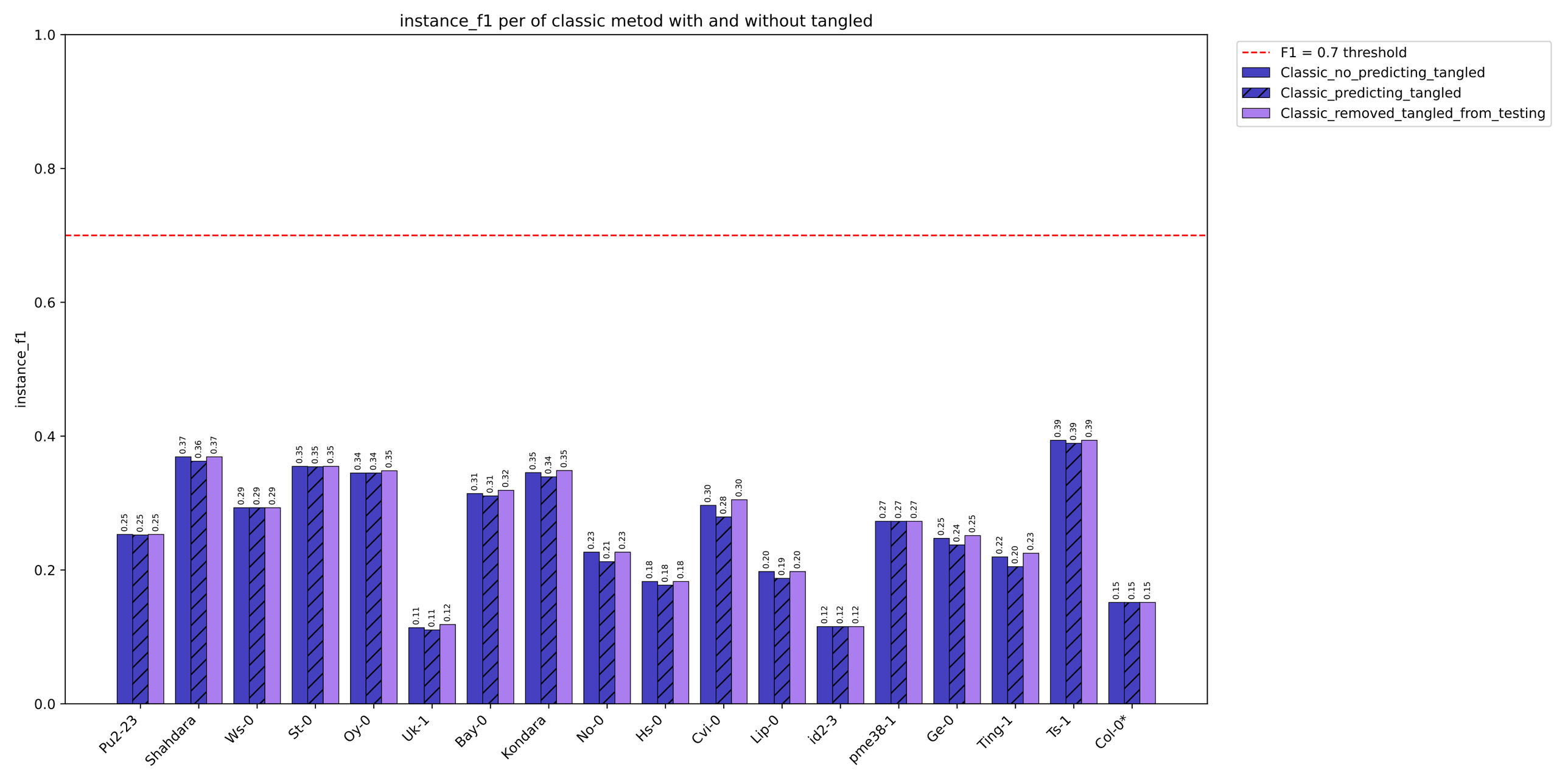

instance_f1 Classic (tangled) with and without inside root hairs
F1 = 0.7 threshold
Classic
Classic not counting inside
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.11
0.12
0.12
0.13
0.15
0.17
0.19
0.20
0.18
0.20
0.20
0.22
0.21
0.22
0.24
0.25
0.25
0.27
0.25
0.27
0.28
0.29
0.27
0.30
0.30
0.31
0.29
0.33
0.31
0.34
0.34
0.36
0.35
0.38
0.34
0.38
0.36
0.39
0.39
0.41
Uk-1
id2-3
Col-0*
Lip-0
Hs-0
Ting-1
No-0
Ge-0
Pla-0
Pu2-23
Cvi-0
pme38-1
Sav-0
Ws-0
Bay-0
Kondara
St-0
Oy-0
Shahdara
Ts-1

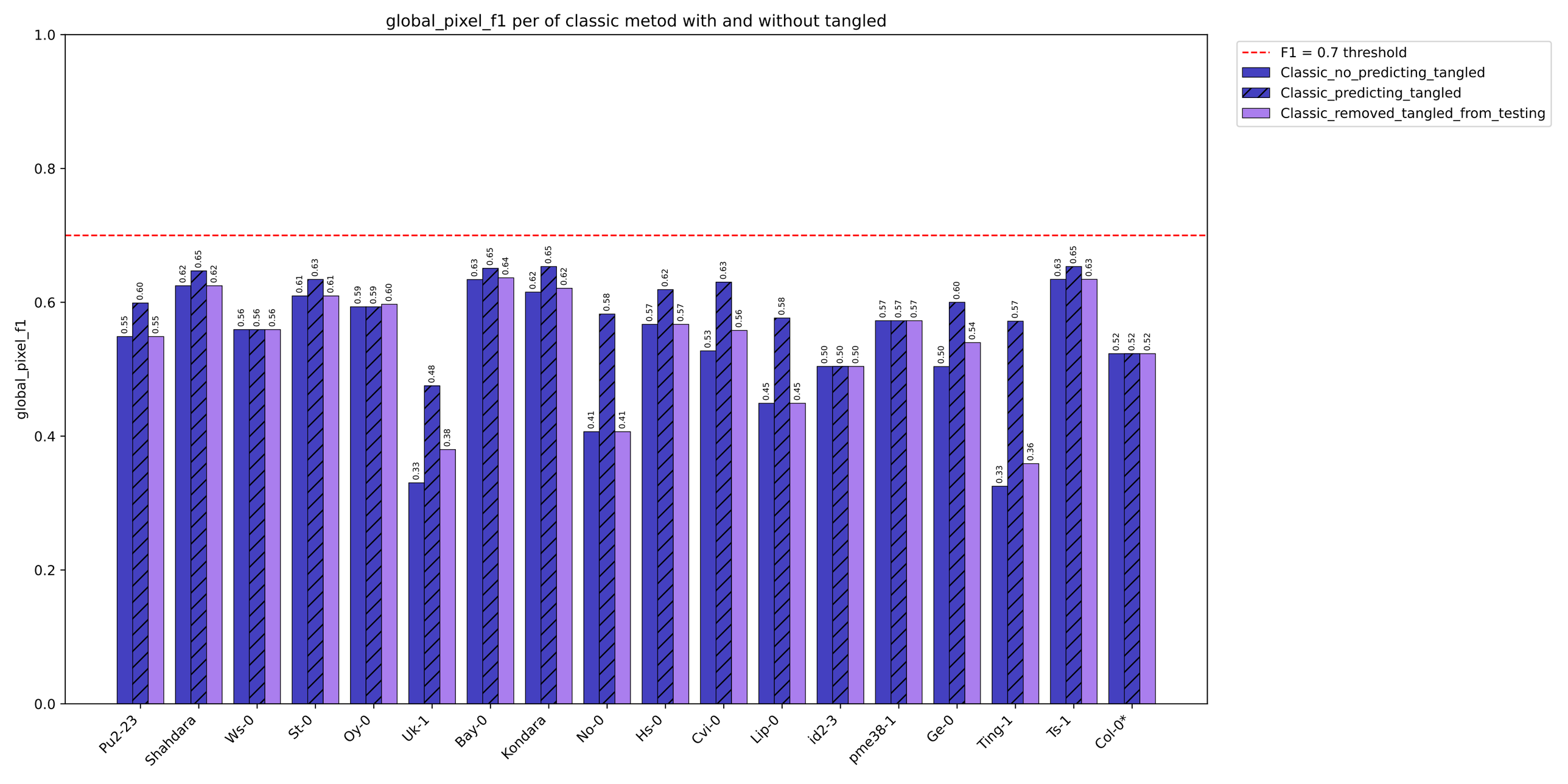

instance_f1 Classic (no tangled) with and without inside root hairs
F1 = 0.7 threshold
Classic
Classic not counting inside
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
0.11 0.12
0.12 0.13
0.15 0.17
0.18 0.21
0.20 0.21
0.22 0.24
0.23 0.24
0.25 0.27
0.25 0.27
0.26 0.28
0.27 0.30
0.30 0.31
0.31 0.32
0.29 0.33
0.31 0.35
0.35 0.36
0.35 0.38
0.34 0.38
0.37 0.40
0.39 0.42
Uk-1
id2-3
Col-0*
Hs-0
Lip-0
Ting-1
No-0
Ge-0
Pu2-23
Pla-0
pme38-1
Cvi-0
Sav-0
Ws-0
Bay-0
Kondara
St-0
Oy-0
Shahdara
Ts-1


global_pixel_f1 per Genotype with and without inside root hairs
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
Classic_yes_tangled
Classic_no_inside_yes_tangled
Classic_no_tangled
classic_no_inside_no_tangled
Pu2-23 0.60 0.60 0.55 0.55
Shahdara 0.65 0.65 0.62 0.63
Ws-0 0.56 0.57 0.56 0.57
St-0 0.63 0.64 0.61 0.61
Oy-0 0.59 0.60 0.59 0.60
Uk-1 0.48 0.48 0.33 0.33
Bay-0 0.65 0.66 0.63 0.64
Kondara 0.65 0.66 0.62 0.62
No-0 0.58 0.58 0.41 0.41
Hs-0 0.62 0.63 0.57 0.58
Cvi-0 0.63 0.63 0.53 0.53
Lip-0 0.58 0.58 0.45 0.45
id2-3 0.50 0.51 0.50 0.51
pme38-1 0.57 0.58 0.57 0.58
Ge-0 0.60 0.61 0.50 0.51
Ting-1 0.57 0.58 0.33 0.33
Ts-1 0.65 0.66 0.63 0.64
Col-0* 0.52 0.53 0.52 0.53
Sav-0 0.59 0.60 0.47 0.47
Pla-0 0.58 0.58 0.45 0.46


instance_f1 per Genotype with and without inside root hairs
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
Classic_yes_tangled
Classic_no_inside_yes_tangled
Classic_no_tangled
classic_no_inside_no_tangled
Pu2-23 0.25 0.27 0.25 0.27
Shahdara 0.36 0.39 0.37 0.40
Ws-0 0.29 0.33 0.29 0.33
St-0 0.35 0.38 0.35 0.38
Oy-0 0.34 0.38 0.34 0.38
Uk-1 0.11 0.12 0.11 0.12
Bay-0 0.31 0.34 0.31 0.35
Kondara 0.34 0.36 0.35 0.36
No-0 0.21 0.22 0.23 0.24
Hs-0 0.18 0.20 0.18 0.21
Cvi-0 0.28 0.29 0.30 0.31
Lip-0 0.19 0.20 0.20 0.21
id2-3 0.12 0.13 0.12 0.13
pme38-1 0.27 0.30 0.27 0.30
Ge-0 0.24 0.25 0.25 0.27
Ting-1 0.20 0.22 0.22 0.24
Ts-1 0.39 0.41 0.39 0.42
Col-0* 0.15 0.17 0.15 0.17
Sav-0 0.30 0.31 0.31 0.32
Pla-0 0.25 0.27 0.26 0.28


Instance F1 by Genotype (Kruskal-Wallis: H=35.54, p=0.0053)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
Pu2-23
Ws-0
Shahdara
St-0
Uk-1
Oy-0
Kondara
Bay-0
Hs-0
Cvi-0
No-0
Ts-1
Lip-0
Ting-1
Ge-0
id2-3
pme38-1
Col-0*
genotype


instance_f1 per of classic metod with and without tangled
instance_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
Classic_no_predicting_tangled
Classic_predicting_tangled
Classic_removed_tangled_from_testing
Pu2-23
Shahdara
Ws-0
St-0
Oy-0
Uk-1
Bay-0
Kondara
No-0
Hs-0
Cvi-0
Lip-0
id2-3
pme38-1
Ge-0
Ting-1
Ts-1
Col-0*
0.25 0.25 0.25
0.37 0.36 0.37
0.29 0.29 0.29
0.35 0.35 0.35
0.34 0.34 0.35
0.11 0.11 0.12
0.31 0.31 0.32
0.35 0.34 0.35
0.23 0.21 0.23
0.18 0.18 0.18
0.30 0.28 0.30
0.20 0.19 0.20
0.12 0.12 0.12
0.27 0.27 0.27
0.25 0.24 0.25
0.22 0.20 0.23
0.39 0.39 0.39
0.15 0.15 0.15


global_pixel_f1 per of classic metod with and without tangled
global_pixel_f1
1.0
0.8
0.6
0.4
0.2
0.0
F1 = 0.7 threshold
Classic_no_predicting_tangled
Classic_predicting_tangled
Classic_removed_tangled_from_testing
0.55 0.60 0.55
0.62 0.65 0.62
0.56 0.56 0.56
0.61 0.63 0.61
0.59 0.59 0.60
0.33 0.48 0.38
0.63 0.65 0.64
0.62 0.65 0.62
0.41 0.58 0.41
0.57 0.62 0.57
0.53 0.63 0.56
0.45 0.58 0.45
0.50 0.50 0.50
0.57 0.57 0.57
0.50 0.60 0.54
0.33 0.57 0.36
0.63 0.65 0.63
0.52 0.52 0.52
Pu2-23
Shahdara
Ws-0
St-0
Oy-0
Uk-1
Bay-0
Kondara
No-0
Hs-0
Cvi-0
Lip-0
id2-3
pme38-1
Ge-0
Ting-1
Ts-1
Col-0*